

FreshRetailNet-50K

A bounded study of evidence-gated replenishment

Detailed results, scientific figures, audit evidence and a descriptive comparison with M5

Business-school research report · 8 October 2026

30 selected series · 360 numerical runs · 32 live-agent runs · 1,140 policy runs

Source population validation: 50,000 series. Counterfactual cost and waste remain simulated; naturally lost demand has no released ground truth.

Separate exploratory transfer: 480 additional runs; training-only cap calibration after the primary failure

Executive findings

The FreshRetailNet study is complete on a frozen panel of 30 store-product series. It adds 360 conventional replenishment runs, 32 matched parser and LLM runs, and 1,140 independent policy experiments to the existing M5 evidence. Each main simulation run uses all 30 series. The source contains 50,000 series, so this panel covers 0.06% of the release. It supports a bounded engineering conclusion rather than a population estimate.

The original gate did not transfer successfully. Its inherited autonomous spending cap was 1,500 cost-index points, while a clean proposed FreshRetailNet order could require about 1,656 points. In the primary normal parser condition, 14 of 14 decision days were held. This failure is reported before any repair: an architecture can be computationally feasible and still be commercially inactive if its threshold is carried into a new quantity scale without checking.

The central attribution check found identical cost and fill in 16 of 16 matched parser-LLM scenario and seed pairs. The language model was able to participate in a verified replenishment workflow. The tested supplier grammar did not, however, demonstrate an additional economic benefit from choosing an LLM over a capable parser.

Under an injected inventory-field failure, verified record repair changed mean LLM-arm cost from 23,331.29 to 23,331.29 (+0.00%) and fill from 45.2% to 45.2% (+0.00 percentage points). The parser received the same repair tools. Any shared improvement belongs to the recovery architecture, rather than to the language model alone.

Fresh retail adds a second problem: a trustworthy stockout record can still leave demand partly unobserved. This study evaluates recovery on artificially hidden sales, forecasts a protected seven-day holdout, and examines perishability inside disclosed simulators. It does not measure true lost demand during natural stockouts, real retailer profit, actual waste savings, or changes in human trust.

The calibrated controller still fills 79.19% of normal simulated demand, compared with 85.03% for conventional B3 over the same 30 seeds. Its lower indexed cost comes with lower service and remaining holds. Calibration improves an unusable gate; it does not demonstrate a dominant LLM policy.

In that exploratory follow-up, the original normal-operation seed pair changed from 45.2% to 80.2% fill, with held days 14 to 3. This is a verified cap-transfer effect, not an incremental LLM advantage: parser-LLM cost and fill are equal in 240/240 follow-up pairs.

A separately registered follow-up used a training-only spending-cap formula and 30 simulation seeds per cell, producing 480 additional runs. It is exploratory because the fixed-gate failure motivated it after primary observations, and it reuses the holdout. Its complete outputs appear separately; none replaces the primary failure.

1. Study scope, timing and evidence boundaries

The protocol was registered after the request at 2026-10-08T09:19:34+00:00. The experiment cutoff was 2026-10-08T13:25:00+00:00, leaving reporting and publication time before 2026-10-08T14:19:34+00:00. The experiment was frozen at 2026-10-08T09:43:01.674587+00:00. Panel and model selection used training data only. This report consumes the saved outputs and does not restart an experiment.

The evidence has four distinct meanings. Observed fields are released sales, identifiers and hourly stockout annotations. Derived fields include historical censoring summaries and source validation. Inferred fields are recovered demand estimates and forecasts. Simulated fields include stock balances, purchasing, suppliers, lead times, expiry and alternative-policy outcomes. A correctly recorded stockout is therefore different from a deliberately corrupted inventory record.

For compatibility with inherited integer-order tools, the core simulator represents quantities as 100 times the published globally normalized sales amount. These are scaled model quantities, not kilograms, cartons or original physical units. Its legacy schemas contain USD tags, but all FreshRetailNet results are reported as a simulated cost index. The independent age-aware experiment uses continuous normalized quantities and its own dimensionless economics.

Completed scope

Experiment	Series	Seeds	Days	Runs / decisions
Core numerical policies	30	30	7	360 / 2520
Matched parser/LLM factorial	30	2	7	32 / 224
Age-aware sensitivity grid	30	30	7	1,080 runs
Stockout-gate mechanism ablation	30	30	7	60 runs
Forecast final holdout	30	Not simulation seeds	7	210 targets × 4 methods

2. Source validation and computational panel

The pinned dataset revision is 08c1fab7f9257bc73679d415d65d644165d351d4. Source checks covered 4,500,000 training rows, 50,000 store-product series, 865 products, 898 stores and 18 cities. The downloaded files contain 865 distinct product IDs, resolving the card and paper counting discrepancy for this revision. Daily sales agree with the sum of hourly sales within $7.11e-15$. Duplicate keys, hierarchy conflicts, invalid stockout encodings and inconsistent stockout counts were absent.

The publisher's implementation and the released arrays confirm that 1 means out of stock. Its 16 operating-hour slots run from 06:00 through 21:00. The training hourly stockout share is 19.877%, while 44.267% of daily rows contain at least one stockout hour. These are different denominators. There are 16,073 training rows with discount values outside $[0,1]$; the sales records were retained and discount was excluded from forecasting features.

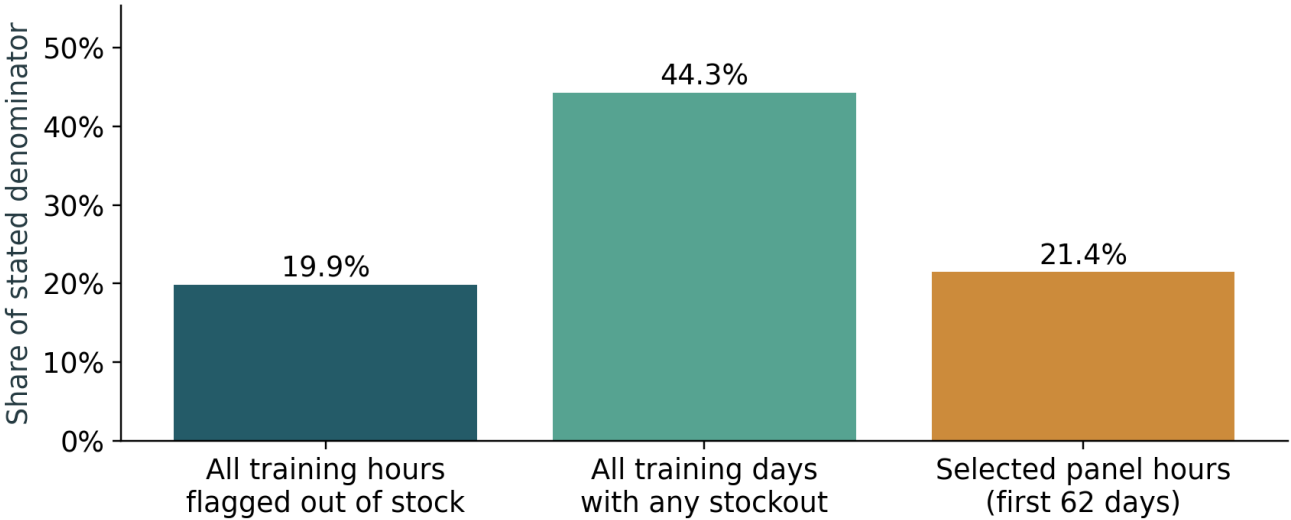
Panel selection used only the first 62 training days. Eligible series needed 62 complete daily rows and mean normalized daily sales between 0.02 and 1.0. Of 37,097 eligible series, three equal-rank historical censoring strata were formed. Ten series were then chosen by deterministic SHA-256 ranking from each stratum. This balances historical censoring exposure for a computational experiment; it does not produce a representative sample of stores or products.

After the forecast selection receipt was frozen, evaluation integrity checks examined all 350,000 released rows. They retain all 50,000 series for seven days, with no duplicate keys or stock-flag alignment failures. Thus the complete source check covers 4,850,000 rows. Holdout profiling does not change the training-selected modeling panel.

Selected panel by first 62-day censoring stratum

Stratum	Series	Mean hourly stockout share	Mean normalized sales
0	10	12.43%	0.6318
1	10	18.31%	0.6352
2	10	33.61%	0.4843

Historical availability is observable; lost demand is not



Training population rates use 72 million operating-hour flags and 4.5 million daily rows. The selected-panel rate uses its first 62 training days. Daily and hourly percentages have different denominators. None measures the quantity of naturally lost demand.

3. F1: demand recovery where a target is identifiable

Availability flags tell us when demand could have been hidden, but the public data do not tell us how much was lost. The identifiable test therefore hides contiguous blocks of 2, 4 or 8 operating hours on originally fully stocked validation days. The hidden values provide a measurable target. The original daily total is also withheld from reconstruction, so it cannot reveal the sum of the hidden observations.

The test compares no recovery, an in-stock hourly profile, and a historical LightGBM regression. Models are fitted before each validation origin. The regression won the registered hidden-hour MAE criterion: 0.04230 against 0.04302 for zero imputation, a modest 1.67% improvement. Its hidden-hour WAPE was nevertheless 98.33% and signed relative bias was -83.10%. Sparse hourly outcomes make small MAE compatible with substantial under-recovery; this is not evidence that naturally lost demand has been reconstructed accurately.

Aggregation tells a different story. The hourly profile's daily reconstruction WAPE was 19.62%, compared with 29.26% for no recovery, a 32.96% reduction. Its daily signed bias was +3.07%. The learned method's daily WAPE was 25.80% and bias -24.32%. The hourly profile had worse hidden-hour WAPE, at 135.18%, but better daily reconstruction. A planner should therefore inspect the outcome needed for the decision rather than assume that one recovery leaderboard answers every question.

There were 495 originally fully stocked validation days, each masked at three durations: 1,485 dependent mask cases. Each method produced 6,930 hidden-hour predictions. These repeated masks are not 6,930 independent businesses. Estimated additions on natural stockout hours remain in separate inferred fields; they are not observed customer demand. The methods are point estimators, so no interval calibration claim is made.

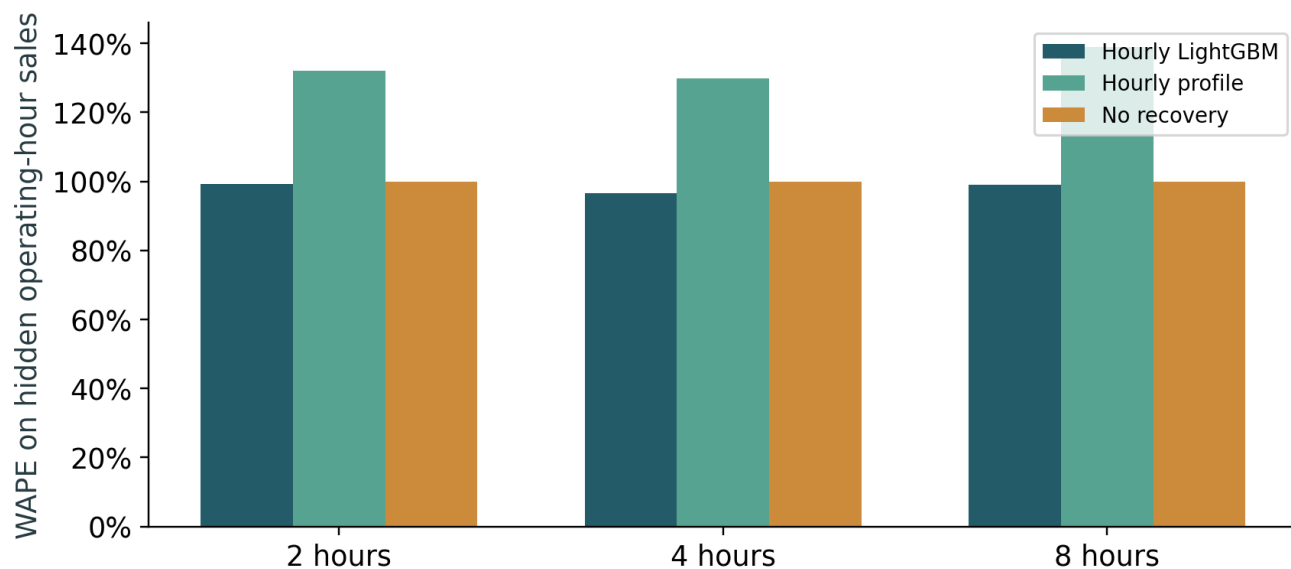
Pooled artificial-mask operating-hour recovery

Method	N predictions	MAE	WAPE	Signed bias
Hourly LightGBM	6930	0.04230	98.33%	-83.10%
Hourly profile	6930	0.05815	135.18%	10.50%
No recovery	6930	0.04302	100.00%	-100.00%

Pooled daily reconstruction from artificial masks

Method	Masked cases	WAPE	Signed bias
Hourly LightGBM	1485	25.80%	-24.32%
Hourly profile	1485	19.62%	3.07%
No recovery	1485	29.26%	-29.26%

Recovery accuracy where a target can actually be checked



Originally observed sales are hidden in 2-, 4- or 8-hour validation blocks. Models fit only earlier data. The target is deliberately withheld sales, rather than unobserved natural stockout demand. Mask cases overlap and are not independent retailers.

4. F2: forecasting censored and fully stocked outcomes

Four expanding validation origins at training day indices 62, 69, 76 and 83 each forecast seven days. Recovery models, profiles and forecasting models are refitted from each allowable prefix. The final models use all 90 training days, after which the official seven-day evaluation split is opened once. Forecasts use historical sales lags, calendar date, day of week and series identity. Future actual sales, availability, weather and promotions are excluded.

Validation selected raw-history LightGBM on fully stocked daily WAPE, at 40.52%. On the untouched final holdout, its corresponding WAPE was 38.84%, against 42.84% for seasonal naive: a 9.33% relative reduction. Learned-recovery LightGBM scored 38.95%, and profile-recovery LightGBM scored 40.07%. These results provide no substantial forecasting advantage from the selected recovery step on the fully stocked holdout.

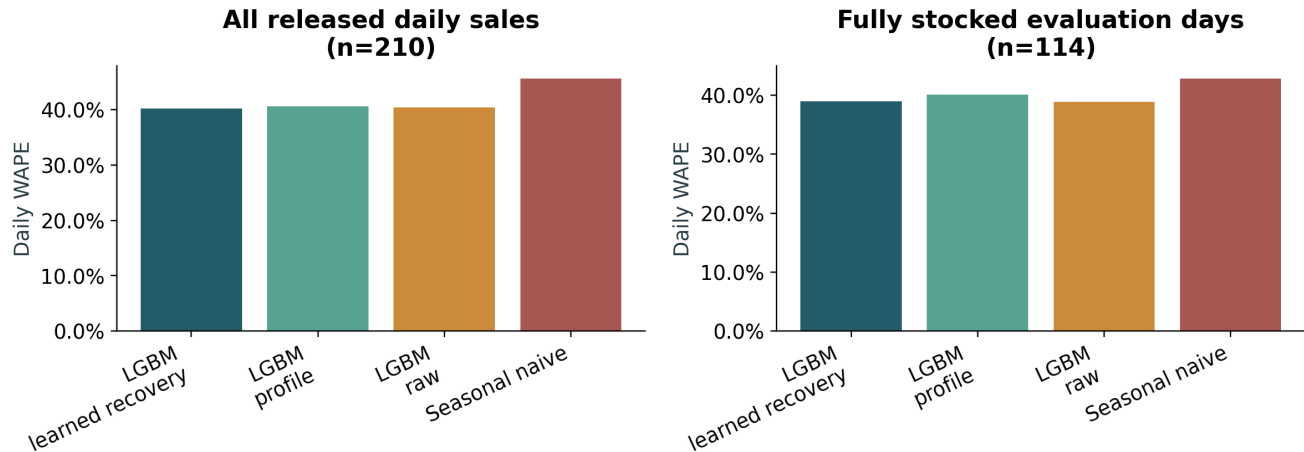
The fully stocked holdout contains 114 of 210 daily observations. Against all recorded sales, WAPE was 40.41% for raw LightGBM, 40.26% for learned-recovery LightGBM, 40.61% for profile recovery and 45.63% for seasonal naive. The small ordering change across scoring subsets matters: all recorded sales can be censored, while fully stocked outcomes are observable but conditionally selected. Neither target reveals natural lost demand.

All 20 leakage checks left forecasts bitwise unchanged after future sales, hourly sales, availability and contextual fields were altered. Final selection was saved before the evaluation read. This verifies the implemented feature boundary; it does not by itself validate the demand and commercial assumptions used later in simulation. No LLM is involved in these recovery or forecast comparisons.

Final seven-day forecast scores

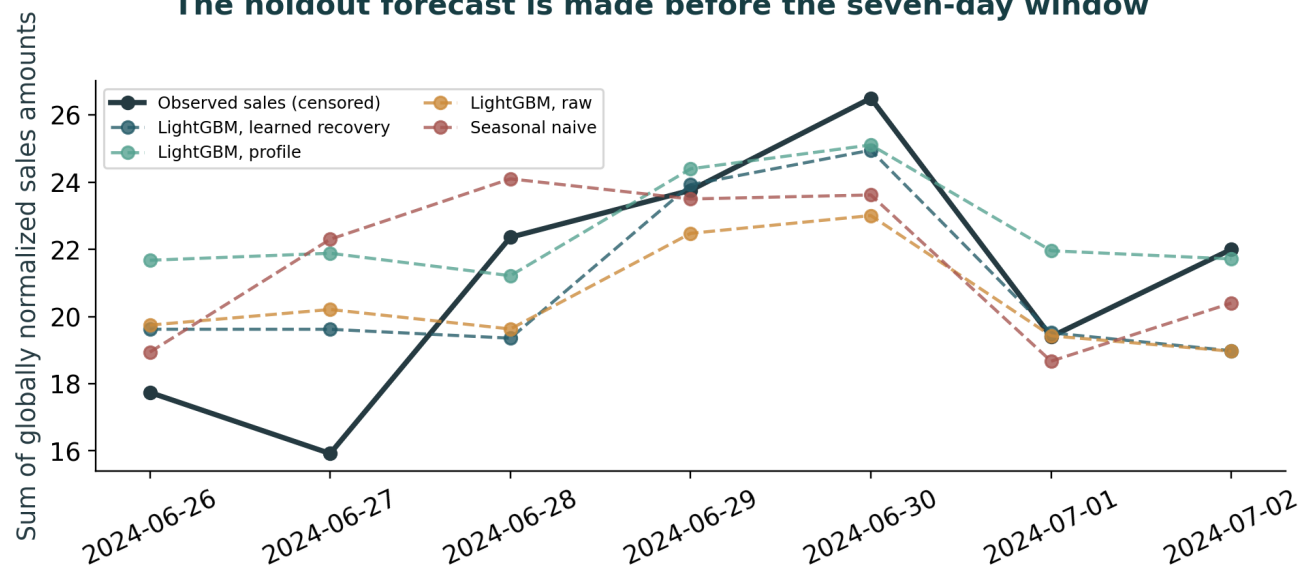
Method	Target	N	WAPE	Bias
LightGBM, learned recovery	All sales	210	40.26%	-1.16%
LightGBM, learned recovery	Fully stocked	114	38.95%	7.11%
LightGBM, profile	All sales	210	40.61%	6.96%
LightGBM, profile	Fully stocked	114	40.07%	11.49%
LightGBM, raw	All sales	210	40.41%	-2.86%
LightGBM, raw	Fully stocked	114	38.84%	5.93%
Seasonal naive	All sales	210	45.63%	2.61%
Seasonal naive	Fully stocked	114	42.84%	13.02%

Forecast quality depends on which sales are observable



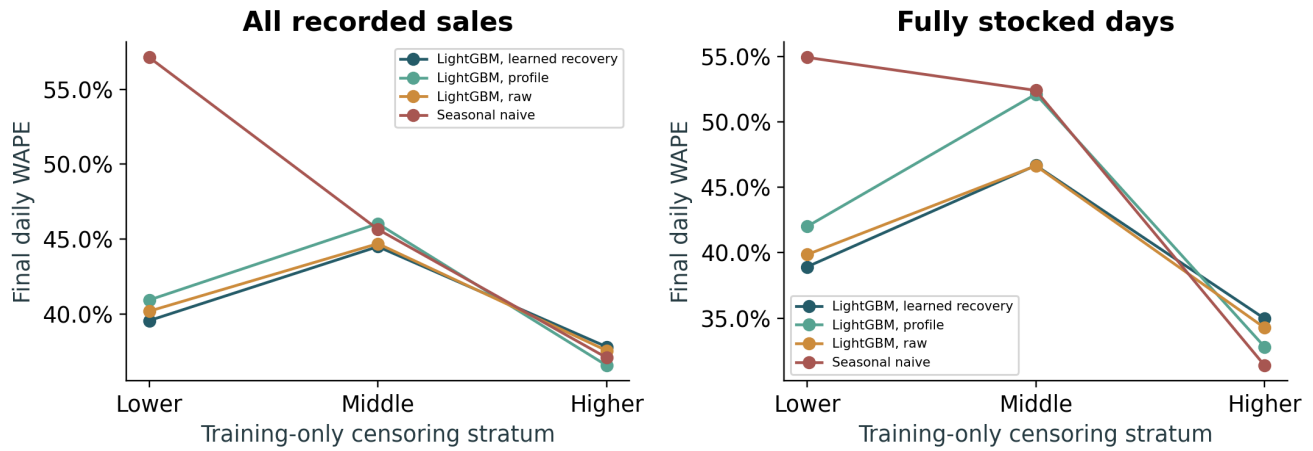
All four frozen forecasts cover the same 30 series and seven-day holdout. All-day sales can be censored; fully stocked days are conditionally observable. Future masks, weather, promotion and actual sales are never forecast features.

The holdout forecast is made before the seven-day window



Panel aggregate predictions are made at the final training origin. Later forecast lags use predictions. Globally normalized sales are neither physical quantities nor a currency.

Historical censoring changes the forecasting problem



Ten series per stratum, defined before evaluation using the first 62 training days. Errors are weighted by the sum of actual scored sales within each group. Evaluation availability selects the fully stocked scoring subset only; it does not change membership or forecasts. These small strata are descriptive.

5. Core numerical replenishment: cost, service and safety

The core policies are B1 seasonal base-stock, B2 LightGBM (s,S), B3 native GRU negative-binomial forecasting with MILP, and B4 the same system with an evidence gate. Each policy and scenario uses 30 simulation seeds, one decision origin, seven days and all 30 series. The opportunity random streams are shared across competing policies rather than being keyed to their names.

The FreshRetailNet physical environment ages stock and serves it by FIFO under a hypothetical three-day shelf life. The inherited MILP is age-unaware: its objective does not optimize age buckets. These are useful controls, but they cannot be described as a full comparison of perishable MILP policies. The independent calculator experiment later examines expiry-aware decisions explicitly.

The cost index combines assumed purchasing, ordering and transfers, holding, shortage penalties and expiry. Opening stock comes from the disclosed warm-up convention and is not charged again. There is no terminal salvage credit. A seven-day result is sensitive to these conventions. Fill, stockouts and expiry are outcomes in an assumed counterfactual world, rather than observed results of a purchase change at Dingdong.

There were 1,269 active-rule violations in committed requested plans from the naive B1/B2 policies. The conventional B3 MILP already had 0 violations, as did the gated agent configurations. This safety difference cannot be credited uniquely to either the LLM or the gate. The physical environment clamps supplier fulfillment; a requested-plan violation is not evidence of negative physical stock.

The violation total counts failed constraint checks across 1,173 committed decision days, rather than that many distinct purchase decisions. A plan can violate more than one rule. The audited family breakdown below shows which checks failed.

The inherited harmful_execution field combines two tests: violating an actual active constraint, or departing sufficiently from a clean-evidence reference plan. The default quantity-distance thresholds are 12 model units and 50% relative difference. A reference-only flag does not establish a realized harmful order, but it remains a diagnostic rather than being discarded. B3 has 380 such flagged action-days out of 630, while its actual constraint violations are zero.

The original gate did not transfer successfully. Its inherited autonomous spending cap was 1,500 cost-index points, while a clean proposed FreshRetailNet order could require about 1,656 points. In the primary normal parser condition, 14 of 14 decision days were held. This failure is reported before any repair: an architecture can be computationally feasible and still be commercially inactive if its threshold is carried into a new quantity scale without checking.

Core numerical means; counts summed across 30 seeds

Scenario	Policy	Cost index	Fill	Stockout	Violations	Held days
Normal	B1	18,290.32	77.6%	28.8%	210	0
Inventory-field fault	B1	18,610.24	78.5%	27.1%	266	0
Stale feed	B1	19,447.74	75.7%	30.0%	209	0
Normal	B2	19,685.44	69.7%	35.6%	190	0
Inventory-field fault	B2	19,172.46	72.9%	32.1%	204	0

Scenario	Policy	Cost index	Fill	Stockout	Violations	Held days
Stale feed	B2	22,377.34	65.3%	39.3%	190	0
Normal	B3	18,794.03	85.0%	20.1%	0	0
Inventory-field fault	B3	19,026.79	85.0%	20.0%	0	0
Stale feed	B3	19,607.02	83.3%	21.7%	0	0
Normal	B4	24,013.14	43.5%	57.7%	0	210
Inventory-field fault	B4	24,013.14	43.5%	57.7%	0	210
Stale feed	B4	24,013.14	43.5%	57.7%	0	210

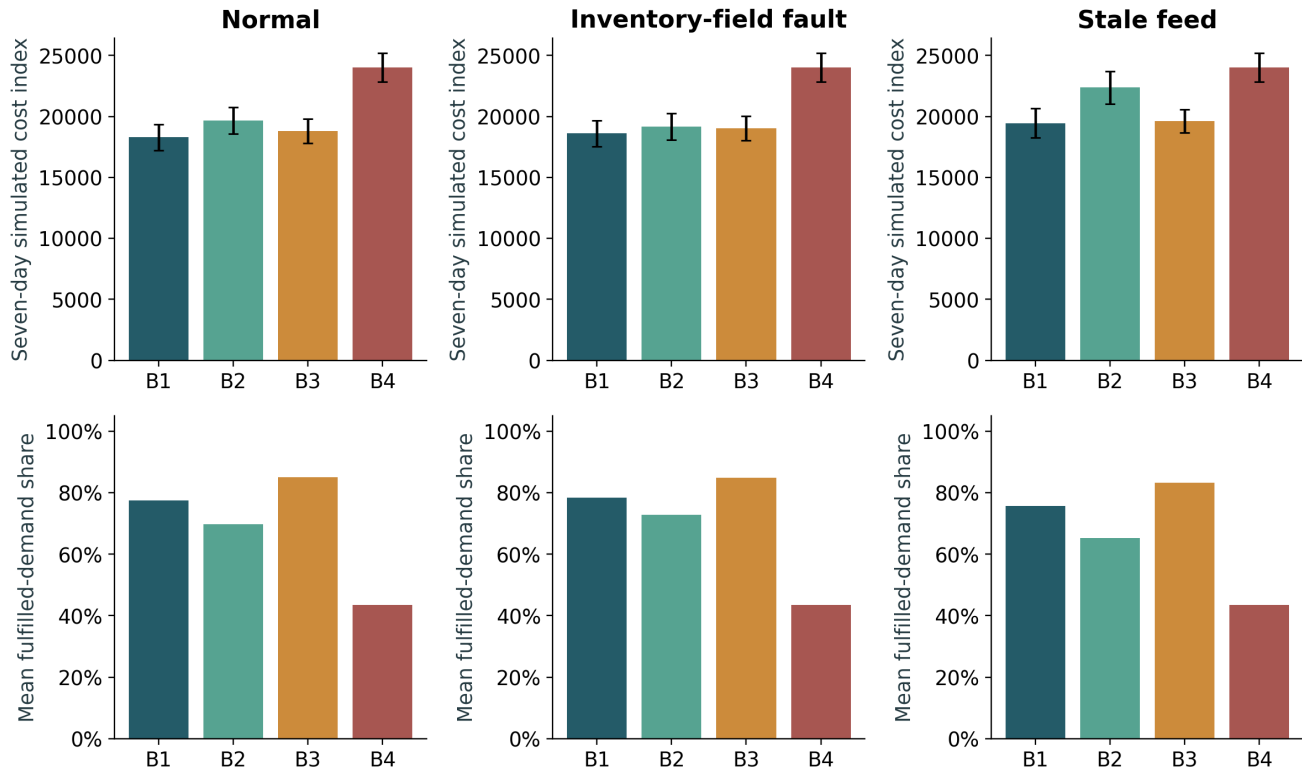
Actual violations and clean-reference flags are different endpoints

Policy	Rule checks violated	Reference-only flagged days	Nonzero action days	Held days
B1	685	7	630	0
B2	584	80	630	0
B3	0	380	630	0
B4	0	0	0	630

Failed active constraint checks in naive requested plans

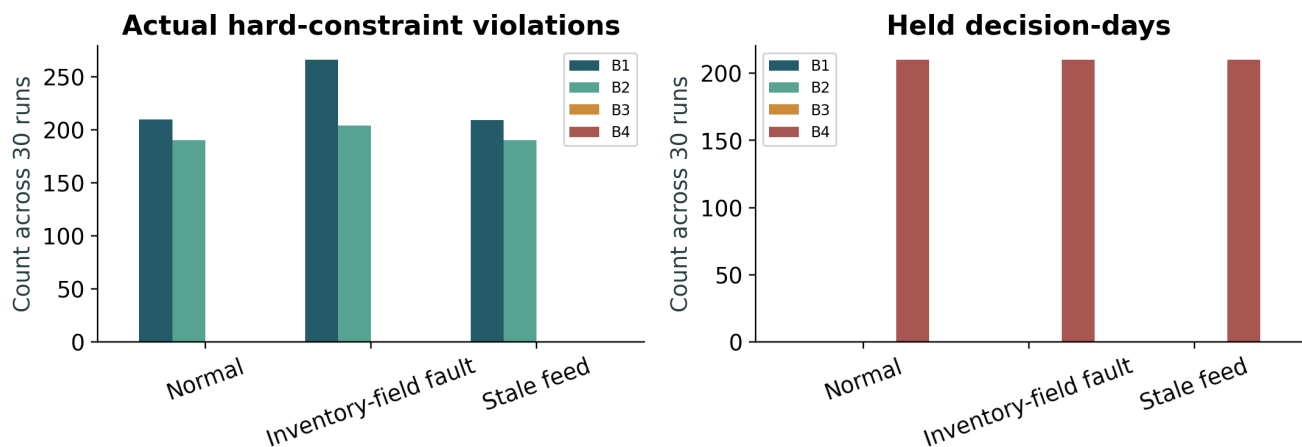
Policy	Supplier capacity	Maximum order	Budget	Total checks
B1	623	14	48	685
B2	550	0	34	584

Conventional policies establish a demanding baseline



Thirty seeds per policy and scenario. Error bars show one standard deviation across conditional simulation seeds. B1: seasonal base-stock; B2: LightGBM (s,S); B3: native GRU-NB with MILP; B4: B3 with an evidence gate. Physical stock expires by FIFO, while the inherited MILP remains age-unaware.

A safe hold has a service cost that should be measured



Counts are summed across 30 runs per cell. Hard violations use actual active constraints and differ from clean-reference deviations. Holds count decision days. A low violation count should be assessed alongside service.

6. Live LLM factorial: contribution and safeguards

The live factorial crosses a capable deterministic parser with local Qwen2.5-1.5B, and verified inventory-record repair off with repair on. It contains 32 runs and 224 decision days across four scenarios and seeds 13 and 29. The model uses Q4_K_M quantization, temperature 0, an 8,192-token context, model seed 42 and three CPU threads. The parser and LLM receive the same source evidence, supplier grammar, forecasts, optimizer and independent checking tools.

The original gate did not transfer successfully. Its inherited autonomous spending cap was 1,500 cost-index points, while a clean proposed FreshRetailNet order could require about 1,656 points. In the primary normal parser condition, 14 of 14 decision days were held. This failure is reported before any repair: an architecture can be computationally feasible and still be commercially inactive if its threshold is carried into a new quantity scale without checking.

Cost and fill were exactly equal in 16/16 matched parser-LLM comparisons. This shows that the model can fit into a bounded replenishment process on the tested grammar. It does not establish an additional economic return from the language layer. Checked source interpretation, bounded execution and refusal to bypass a failed certificate should be evaluated separately from profit or service superiority.

For the inventory-field fault, enabling repair changed the LLM-arm mean cost by +0.00% and fill by +0.00 percentage points. The capable parser has access to the same verified repair. This is record reconciliation: it checks a derived stock field against fresh source and movement evidence. Statistical demand recovery, measured earlier, estimates sales that were not observed; the two operations should not share a single “AI recovery” benefit label.

A semantic mistake can be blocked by an independent verifier before it becomes an unsafe committed order. Accordingly, the audit reports wrong selector choices, attempts, cache imports, holds and actual committed-plan violations separately. Zero committed violations must not be translated into a claim of zero model errors. Reused cached responses are not fresh independent trials.

In the primary live study, 24 physical extraction calls returned 38 exact numerical terms out of 38. That is the bounded numerical head, not the whole supplier corpus. Twelve fresh recovery selections were instruction-compliant in six cases; six wrong selections were refused by independent checks. Across 224 agent decisions, 188 attempted source grounding and all 188 source sets were exact. Each set has 163 clauses, yielding 30,644 checked active clauses; 36 state-gated decisions did not attempt grounding. Exact conditional source coverage must not be relabelled unconditional model accuracy.

The primary agent configurations made 40 nonzero action-days and held 184 decision-days in total. The nonzero actions occur under capacity cuts, whose smaller orders can pass the inherited spending cap. Normal, field-fault and stale-feed scenarios are fully held. These successful bounded actions demonstrate feasibility in a narrow condition, while the normal-operation failure motivates the explicitly separate transfer follow-up.

All 40 primary nonzero agent action-days carry a reference-distance flag, with zero actual constraint violations. The legacy composite therefore labels them harmful_execution even though the flag arises from plan distance alone. The report retains those counts and makes no claim of zero modeled risk or zero disagreement with the reference. Feasibility, clean-reference agreement and achieved service are distinct endpoints.

A concrete simulator case shows both the value and the limit of this design. On decision day 92, the damaged derived inventory for FRN_0422@S0173 was 0, while fresh source and movement balance both showed 444. The model selected the permitted reconciliation tool. Independent checks restored

444 scaled model units, changed 27 permitted quantities and removed the integrity hard failure. The resulting order was still held: proposed spending of 1,657.60 exceeded the 1,500 autonomous cap. Correct exception handling can repair evidence without making the operating policy usable.

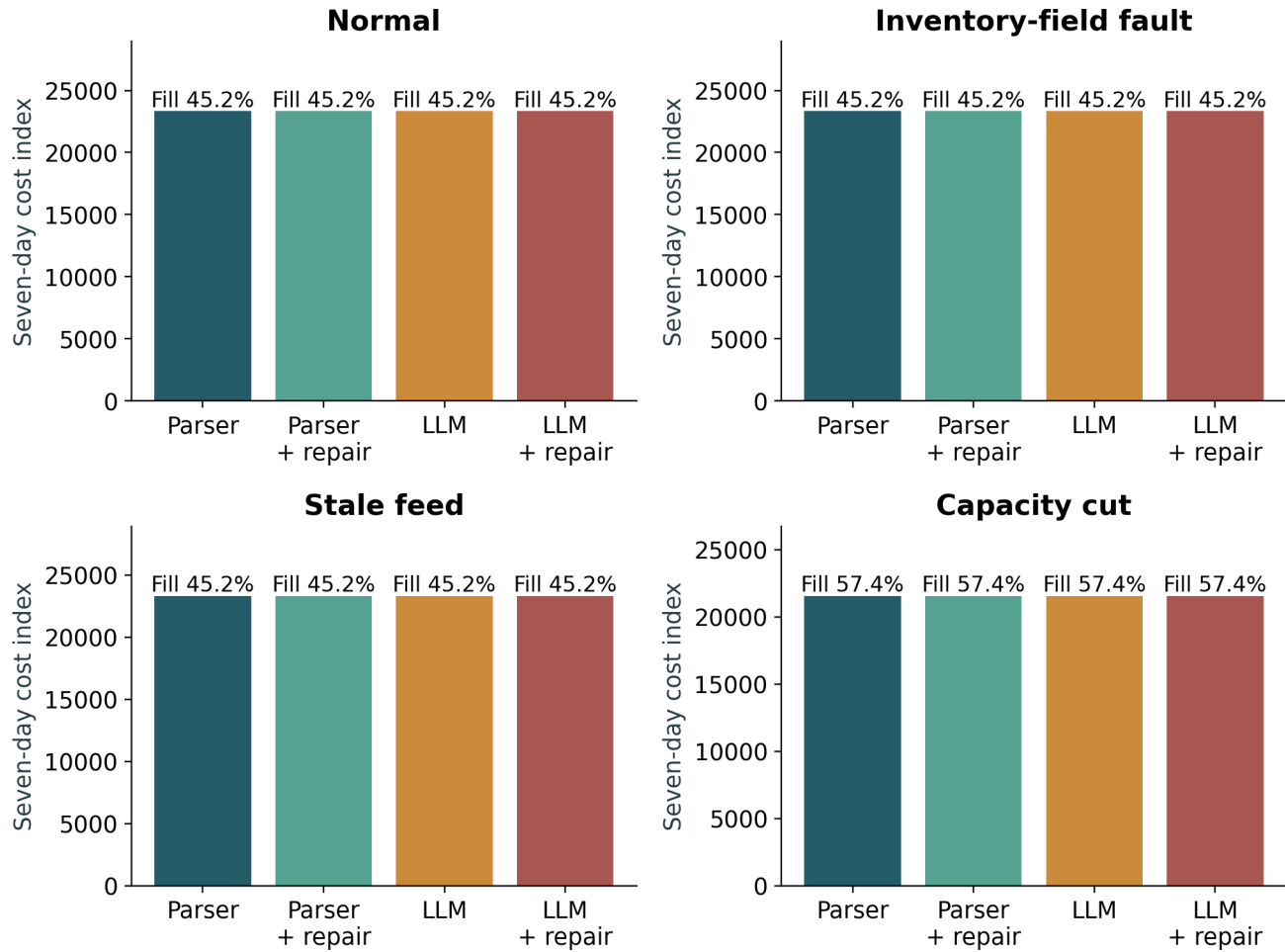
Agent factorial means; two seeds per row

Arm	Scenario	Cost index	Fill	Held days	Violations	Freshcalls
Parser	Normal	23,331.29	45.2%	14	0	0
Parser	Inventory-field fault	23,331.29	45.2%	14	0	0
Parser	Stale feed	23,331.29	45.2%	14	0	0
Parser	Capacity cut	21,572.97	57.4%	4	0	0
Parser + repair	Normal	23,331.29	45.2%	14	0	0
Parser + repair	Inventory-field fault	23,331.29	45.2%	14	0	0
Parser + repair	Stale feed	23,331.29	45.2%	14	0	0
Parser + repair	Capacity cut	21,572.97	57.4%	4	0	0
LLM	Normal	23,331.29	45.2%	14	0	7
LLM	Inventory-field fault	23,331.29	45.2%	14	0	0
LLM	Stale feed	23,331.29	45.2%	14	0	0
LLM	Capacity cut	21,572.97	57.4%	4	0	5
LLM + repair	Normal	23,331.29	45.2%	14	0	7
LLM + repair	Inventory-field fault	23,331.29	45.2%	14	0	6
LLM + repair	Stale feed	23,331.29	45.2%	14	0	6
LLM + repair	Capacity cut	21,572.97	57.4%	4	0	5

Primary execution and reference-distance flags

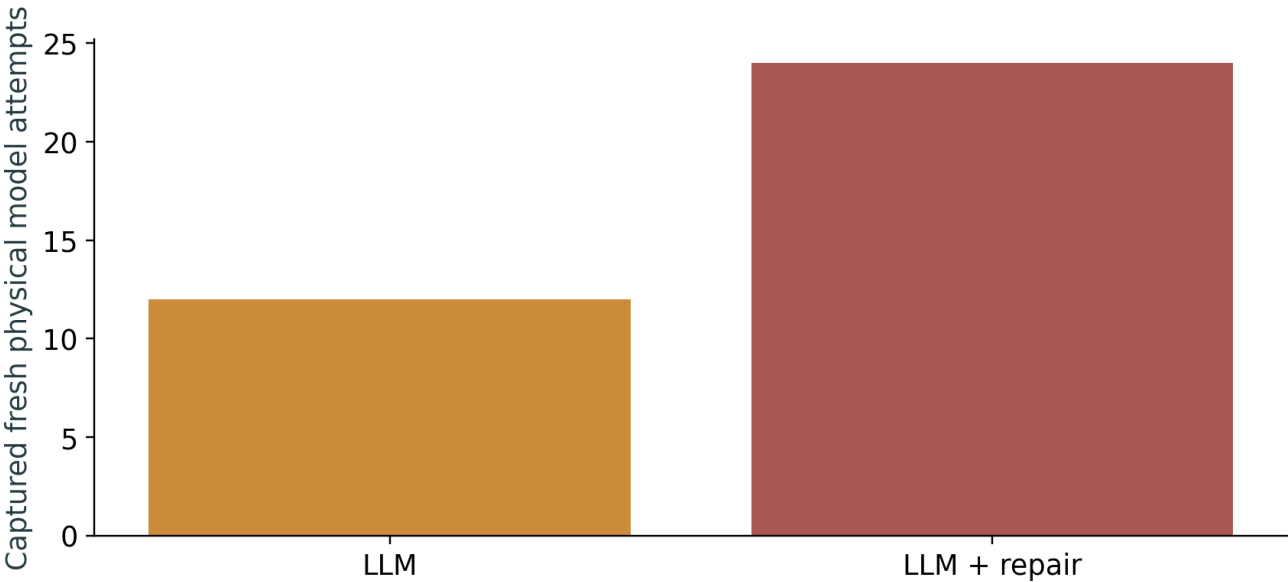
Arm	True rule checks violated	Reference-only flagged days	Nonzero action days	Held days
LLM	0	10	10	46
LLM + repair	0	10	10	46
Parser	0	10	10	46
Parser + repair	0	10	10	46

The LLM shares the same forecasts, tools and safeguards



Four matched controller/repair configurations, four scenarios and two seeds, each using seven days. Repair is verified inventory-record reconciliation. It is not statistical recovery of censored customer demand. All configurations share the numerical tools and source evidence.

Inference is an additional operating burden



Primary registered study physical attempts from the independent audit. The exploratory follow-up has separate counts. Cache imports and typed extraction outputs are not additional calls. Local API charges are zero; CPU hosting and oversight costs are unpriced.

6a. Exploratory spending-cap portability follow-up

The original gate did not transfer successfully. Its inherited autonomous spending cap was 1,500 cost-index points, while a clean proposed FreshRetailNet order could require about 1,656 points. In the primary normal parser condition, 14 of 14 decision days were held. This failure is reported before any repair: an architecture can be computationally feasible and still be commercially inactive if its threshold is carried into a new quantity scale without checking.

The failure led to a separately registered follow-up. Its hypothesis was chosen after the primary study had started; the official holdout had already been used. The study therefore labels all subsequent transfer results exploratory. The original fixed-gate outputs are unchanged and remain in the report and archive.

Cap calibration uses seven earlier historical dates, training a separate GRU through day 75 and simulating June 12–18 (indices 76–82). The rule was fixed before those proposals were generated: $\min(0.8 \times \text{budget}, \text{ceil}(1.10 \times \text{the seven-proposal 95th percentile} / 50) \times 50)$. The historical 95th percentile was 1,659.635; the selected cap was 1,850, below the 2,400 maximum. The independent two-person threshold remains 2,500. All seven proposals were included, including 2 days held by the unchanged extra-spend risk rule. Proposal values, dates, model identity and selection timestamps are saved.

The transfer experiment contains 480 runs and 3,360 decision days: four configurations, four scenarios and 30 seeds. It uses the primary saved GRU tensors and the same simulator, supplier sources, LLM, prompts, constraints and checking tools. Only the spending cap and its version change semantically; output locations, cache and spend ledger are separately named for delivery and audit.

Cost and fill were identical in 240/240 parser-LLM pairs. Under the inventory-field fault, verified repair changed mean LLM cost by -11.71% and fill by +12.25 percentage points. These results distinguish fixing an unusable operating threshold from the value of the language model. They are not an untouched confirmation, and any shared repair benefit remains attributable to the architecture.

For a business, this is an implementation lesson as much as a modeling result. A limit calibrated for unit sales in one retailer can produce systematic holds after quantity normalization changes. Conservative caps remain valuable, but the business must verify that legitimate orders can pass them. That review should use historical proposals and explicit risk budgets, rather than repeatedly adjusting limits until a favourable holdout result appears.

The paired seed results show the benefit and its limits. Under the inventory-field fault, indexed cost falls in 26/30 seeds and rises in 4; fill improves in 30/30. Verified repair restores the normal-case economic and service results for every seed. The stale-feed condition is unchanged because fresh source evidence is unavailable. These are conditional simulator effects; the same recovery benefit occurs with the capable parser.

Calibration restores useful execution, but the remaining gate still carries an opportunity cost. Over the same 30 seeds, normal-operation fill is 79.19% for the calibrated controller and 85.03% for conventional B3, a -5.83-point change. Mean indexed cost is 18,100.60 against 18,794.03 (-3.69%), with 57 held days against 0. Both use the same saved forecaster, simulator and numerical tools and both have zero actual constraint violations. Lower indexed spending accompanies lower service; these values do not establish overall LLM superiority. The parser gives the same calibrated result, and this comparison remains exploratory.

Reference-only deviations are retained separately from actual active-rule violations in this follow-up as well. The legacy harmful_execution composite includes either endpoint. Its label alone cannot identify

realized injury, waste or economic harm; outcomes and the underlying reason for each flag must be inspected.

The independent follow-up audit verifies 480 runs and 3,360 physical decision records. It captures 204 fresh attempts and 308,841 tokens, with 0 client/schema errors. There are 0 actual committed requested-plan violations. Selector compliance remains a separate table entry, so a blocked semantic error is not counted as a successful choice.

The bounded numerical head returns 38/38 exact submitted terms in 24 fresh extraction responses. Separately, 90/180 fresh recovery selections follow the instruction. The 90 stale-feed mis-selections are refused by independent checks. A zero client/schema-error count therefore coexists with substantive reasoning errors; safe final execution depends on the wider control process.

Grounding was attempted on 2,820 of 3,360 follow-up decision days; 2,820 of those attempted source sets were exact. Each available set has 163 clauses, yielding 459,660 exact active-rule records. The other 540 days were state-gated before grounding and remain in the denominator of all decision days. These records combine a bounded model extraction head with deterministic parsing and verification; full-set agreement is not the model's accuracy on every clause.

On the original matched seeds 13 and 29 in normal operation, cap transfer changed mean LLM cost from 23,331.29 to 18,454.95 (-20.90%) and fill from 45.2% to 80.2% (+35.05 points). Held decision days changed from 14 to 3. These paired values isolate the cap change more clearly than comparing the primary two-seed mean with all 30 follow-up seeds, but the question remains exploratory.

Exploratory transfer means; primary results remain separate

Arm	Scenario	Runs	Cost index	Fill	Held days	Violations
LLM	Capacity cut	30	16,942.51	75.9%	15	0
LLM	Inventory-field fault	30	20,500.69	66.9%	105	0
LLM	Stale feed	30	20,500.69	66.9%	105	0
LLM	Normal	30	18,100.60	79.2%	57	0
LLM + repair	Capacity cut	30	16,942.51	75.9%	15	0
LLM + repair	Inventory-field fault	30	18,100.60	79.2%	57	0
LLM + repair	Stale feed	30	20,500.69	66.9%	105	0
LLM + repair	Normal	30	18,100.60	79.2%	57	0
Parser	Capacity cut	30	16,942.51	75.9%	15	0
Parser	Inventory-field fault	30	20,500.69	66.9%	105	0
Parser	Stale feed	30	20,500.69	66.9%	105	0
Parser	Normal	30	18,100.60	79.2%	57	0
Parser + repair	Capacity cut	30	16,942.51	75.9%	15	0
Parser + repair	Inventory-field fault	30	18,100.60	79.2%	57	0
Parser + repair	Stale feed	30	20,500.69	66.9%	105	0

Arm	Scenario	Runs	Cost index	Fill	Held days	Violations
Parser + repair	Normal	30	18,100.60	79.2%	57	0

Training-only proposed spending; all seven days retained

Historical decision	Spending index
76	1,658.700
77	1,657.600
78	1,659.800
79	1,657.050
80	1,649.900
81	1,506.350
82	1,659.250

Remaining clean-operation gate trade-off; 30 paired seeds, exploratory

Controller	Runs	Cost index	Fill	Held days	True violations
Conventional B3 MILP	30	18,794.03	85.03%	0	0
Calibrated LLM with the shared gate	30	18,100.60	79.19%	57	0

Exploratory actual violations versus clean-reference distance

Arm	True rule checks violated	Reference-only flagged days	Nonzero action days	Held days
LLM	0	265	558	282
LLM + repair	0	311	606	234
Parser	0	265	558	282
Parser + repair	0	311	606	234

Independent audit of the exploratory transfer

Measure	Recorded count
Completed runs	480
Decision days	3360
Fresh model attempts	204
Recorded model tokens	308841
Client or schema errors	0

Measure	Recorded count
Violated active-rule checks in requested plans	0
Held decision days	1032
Instruction-compliant recovery selections	90
Incorrect recovery selections	90
Matched parser-LLM pairs	240
Pairs with identical primary outcomes	240
Pairs with identical daily actions	240
Verified physical stock balances	3360
Forbidden model-context findings	0

Exploratory source grounding; conditional coverage is explicit

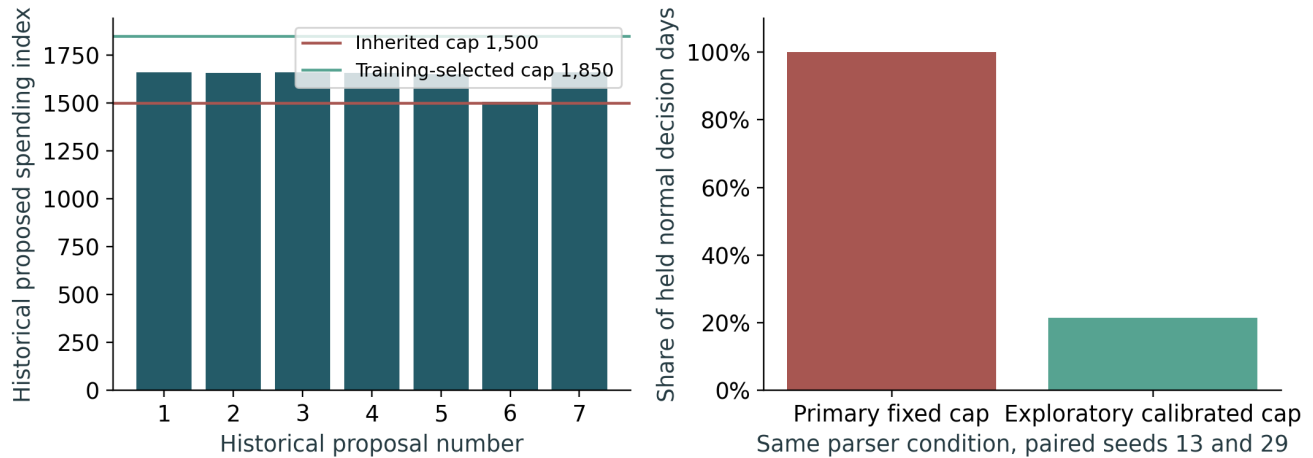
Configuration	Grounding attempted	Exact full source sets	State-gated days	False clauses	Omitted clauses
Parser	660	660	180	0	0
Parser + repair	750	750	90	0	0
LLM	660	660	180	0	0
LLM + repair	750	750	90	0	0

Gate transfer on the same two original seeds; exploratory holdout reuse

Arm	Scenario	Cost change	Fill change (pp)	Held days before/after
LLM	Capacity cut	-21.24%	+18.64	4/1
LLM	Inventory-field fault	-14.97%	+23.62	14/7
LLM	Stale feed	-14.97%	+23.62	14/7
LLM	Normal	-20.90%	+35.05	14/3
LLM + repair	Capacity cut	-21.24%	+18.64	4/1
LLM + repair	Inventory-field fault	-20.90%	+35.05	14/3
LLM + repair	Stale feed	-14.97%	+23.62	14/7
LLM + repair	Normal	-20.90%	+35.05	14/3
Parser	Capacity cut	-21.24%	+18.64	4/1
Parser	Inventory-field fault	-14.97%	+23.62	14/7
Parser	Stale feed	-14.97%	+23.62	14/7
Parser	Normal	-20.90%	+35.05	14/3
Parser + repair	Capacity cut	-21.24%	+18.64	4/1
Parser + repair	Inventory-field fault	-20.90%	+35.05	14/3

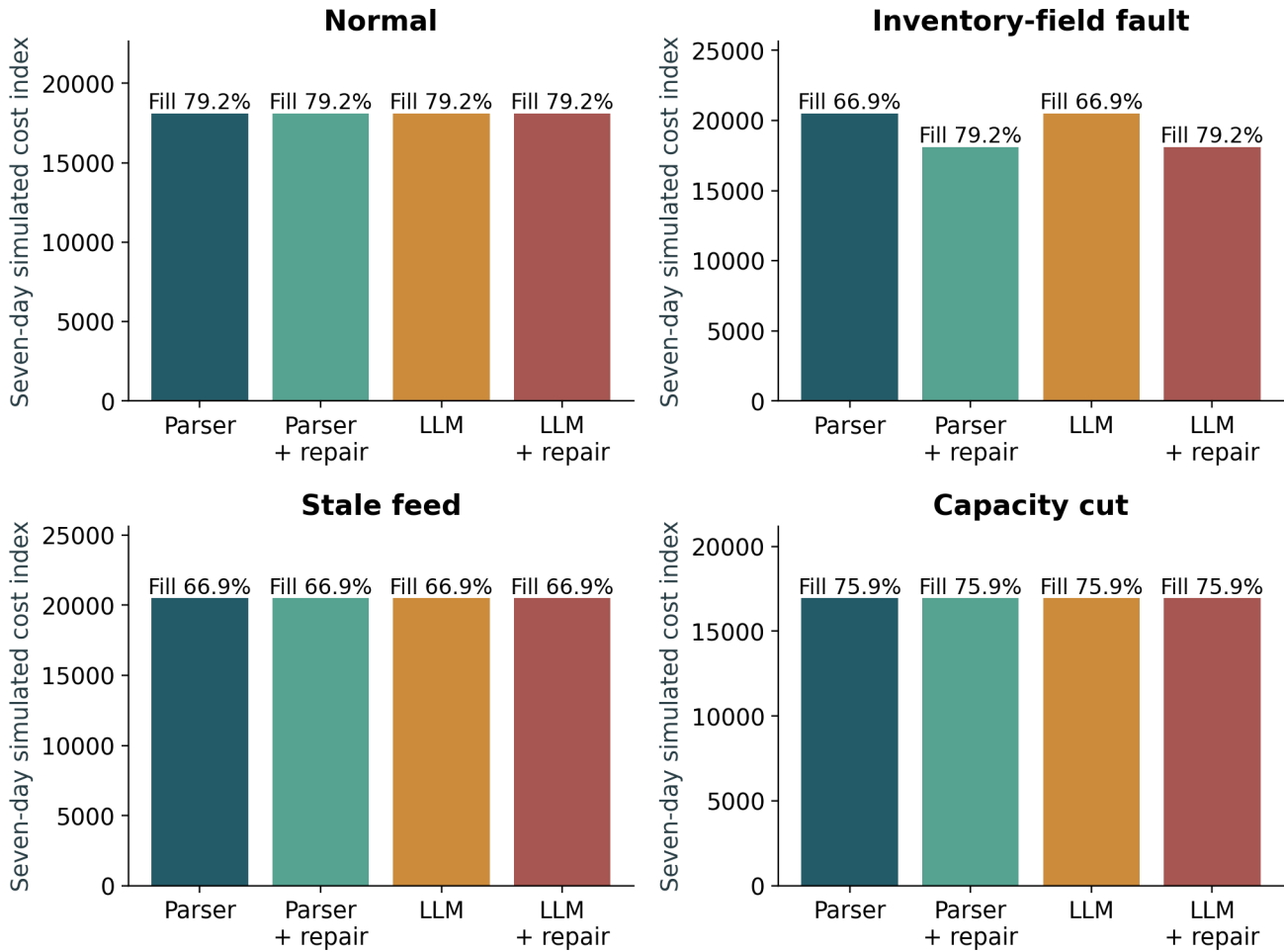
Arm	Scenario	Cost change	Fill change (pp)	Held days before/after
Parser + repair	Stale feed	-14.97%	+23.62	14/7
Parser + repair	Normal	-20.90%	+35.05	14/3

A spending threshold needs calibration to its new scale

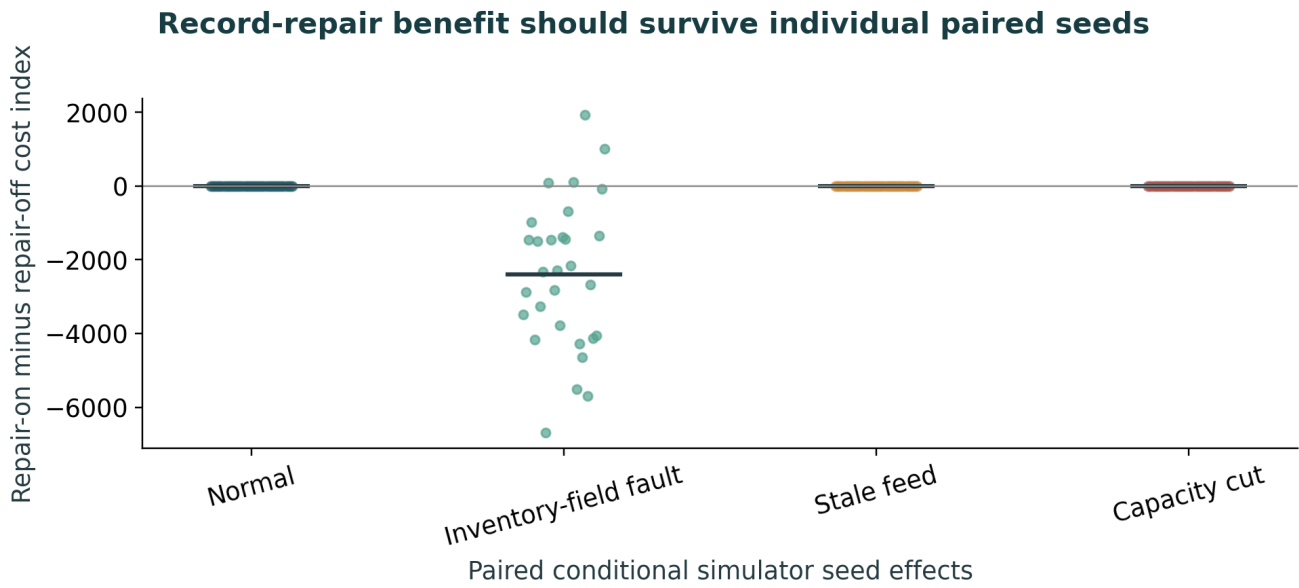


The right panel compares the same original seeds, 13 and 29, with 14 normal parser decision days under each cap. Full 30-seed means appear in the following figure. The fixed-gate failure motivated this exploratory follow-up after primary evaluation began. Seven historical calibration proposals use training-only dates and a separately fitted model. The cap formula was registered before calibration. Reusing the official evaluation window means the follow-up is not confirmatory.

Exploratory calibrated transfer: useful execution and fair attribution



30 paired simulator seeds per configuration and scenario, all 30 series and seven days. The original saved forecast, prompts, model and numerical tools are retained. Only the spending threshold and its version change semantically. This follow-up was chosen after inspecting primary portability and reuses the holdout.



Each point is a matched simulator seed in the exploratory follow-up, with a short line for its mean. Seeds vary a shared assumed environment, not independent retailers or historical forecast origins. The graph makes adverse and null effects visible without a confirmatory population claim.

7. Independent age-aware policy and stockout-gate sensitivity

The independent sensitivity grid crosses raw and validation-selected recovered forecast histories, order-up-to and age-aware calculators, shelf lives of 1, 3 and 7 days, demand multipliers of 1, 1.25 and 1.5, and 30 seeds. This gives 1,080 runs. Another 60 runs compare a censoring-aware gate with an intentionally defective rule that holds each series after a valid prior stockout. At the origin, that flag comes from the final training day; later it comes from the policy's own preceding simulated shortage. The rule therefore tests a feedback mechanism, using no future released mask. It is not the primary strong-parser comparator.

Within each cell, both policies receive the same observed-sales demand proxy multiplied by the same registered scenario factor. A recovered estimate from one competing arm is never treated as ground truth. Orders change counterfactual availability and stockouts. Published stockout flags remain evidence about the original retailer's observations, rather than outcomes that are forced onto every new stocking policy.

This simulator uses continuous normalized quantities, explicit FIFO expiry and inventory conservation. Opening stock is two days of raw historical mean demand, spread across age buckets and supplied as the initial condition. Expected receipts are 98% of ordered quantity; keyed lead-time probabilities are 90%, 8% and 2% for one, two and three days. Unit purchase, shortage and expiry indices are 1, 5 and 1, with holding 0.015 and fixed order cost 0.02 per line. Receipt day counts as a usable life day. Terminal inventory and paid pipeline are recorded without a salvage credit. Where target calculations extend beyond the seven reported days, the frozen forecast repeats by weekday; no later observed outcome is added.

The calculators also differ in target timing. The simple order-up-to rule covers today and tomorrow against current stock and pipeline. The age-aware rule projects stock surviving today's service and expiry, then covers up to two future days subject to shelf life. Their contrast is therefore not a pure isolated effect of adding age information. Their cost coefficients and quantity conventions also differ from the core MILP, so absolute costs are not pooled across experiments.

At the default three-day shelf life and 1.25 demand multiplier, age-aware raw forecasts cost 303.3008 with 92.0804% fill and 7.5433% expired quantity per demand. Learned-recovery forecasts cost 300.5691 with 92.6071% fill and 8.0615% expiry per demand: cost fell about 0.90% and fill rose 0.53 points, while expiry rose 0.52 points. Some other cells increased cost. Recovery is therefore a trade-off, rather than an unconditional improvement.

The same default censoring-aware gate cost 300.5691 with 92.6071% fill and no held series-days. The deliberately naive stockout-hold gate cost 365.9235 with 82.4591% fill and a mean of 56.9333 held series-days. This benefit belongs to interpreting valid availability evidence correctly. It uses no live language inference and cannot be reported as an LLM gain.

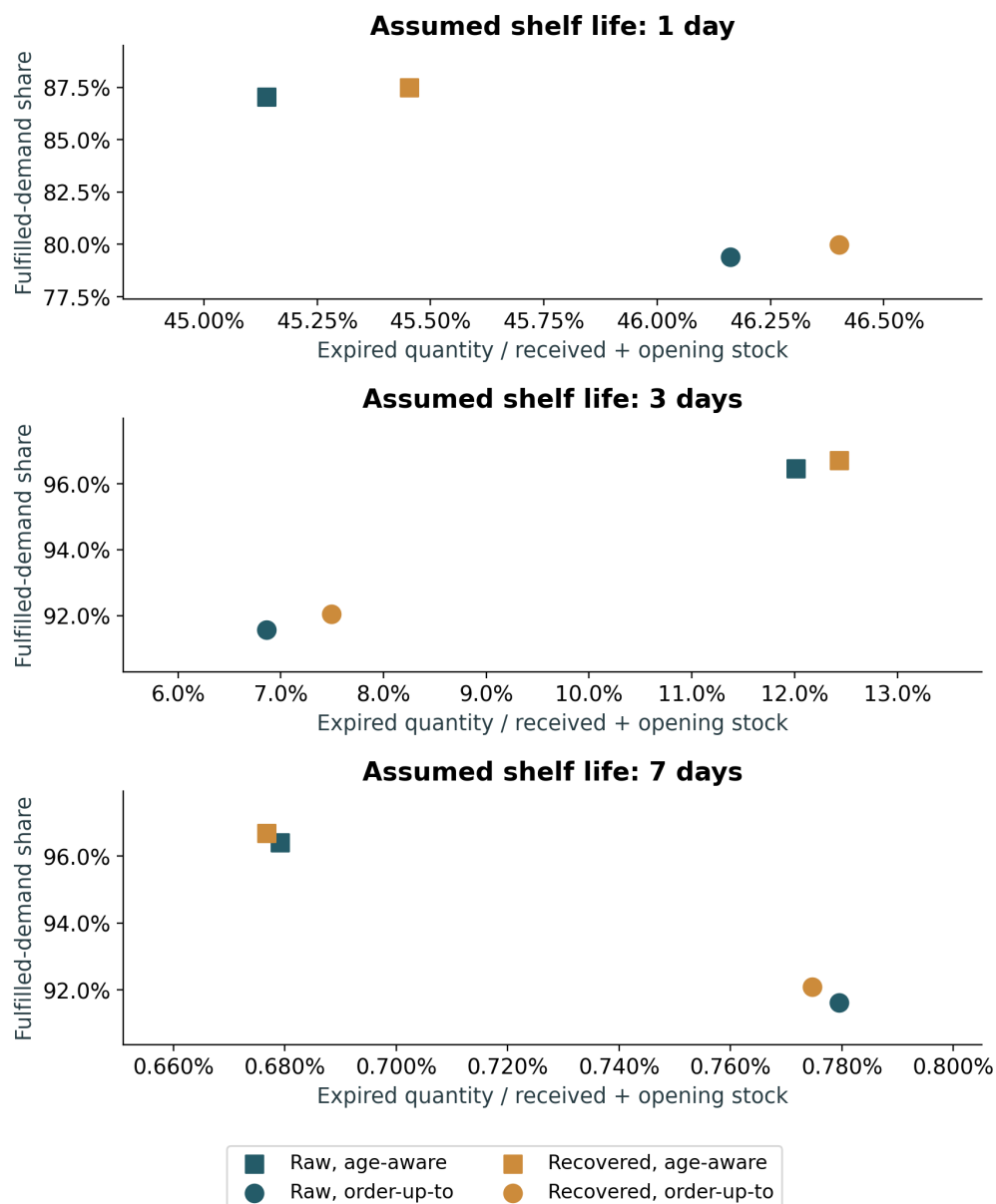
Independent sensitivity means; full grid in CSV

Calculator	History	Life	Demand×	Cost index	Fill	Expiry share
age_aware	raw	1	1.0	444.2409	87.0%	45.1%
age_aware	raw	1	1.25	490.2626	82.1%	35.3%
age_aware	raw	1	1.5	571.9640	76.2%	28.0%
age_aware	raw	3	1.0	250.8000	96.5%	12.0%

Calculator	History	Life	Demandx	Cost index	Fill	Expiry share
age_aware	raw	3	1.25	303.3008	92.1%	6.6%
age_aware	raw	3	1.5	387.9095	86.5%	4.0%
age_aware	raw	7	1.0	208.4383	96.4%	0.7%
age_aware	raw	7	1.25	278.1695	92.1%	0.6%
age_aware	raw	7	1.5	372.4381	86.5%	0.5%
age_aware	selected_recovery	1	1.0	445.3558	87.5%	45.5%
age_aware	selected_recovery	1	1.25	489.5699	82.7%	35.6%
age_aware	selected_recovery	1	1.5	570.7272	76.7%	28.3%
age_aware	selected_recovery	3	1.0	251.1878	96.7%	12.4%
age_aware	selected_recovery	3	1.25	300.5691	92.6%	7.0%
age_aware	selected_recovery	3	1.5	383.2645	87.1%	4.3%
age_aware	selected_recovery	7	1.0	206.3261	96.7%	0.7%
age_aware	selected_recovery	7	1.25	273.6730	92.6%	0.5%
age_aware	selected_recovery	7	1.5	366.4855	87.1%	0.5%
order_up_to	raw	1	1.0	472.5299	79.4%	46.2%
order_up_to	raw	1	1.25	534.9456	74.6%	36.8%
order_up_to	raw	1	1.5	625.4762	69.2%	29.6%
order_up_to	raw	3	1.0	233.4785	91.6%	6.9%
order_up_to	raw	3	1.25	325.7840	84.4%	4.1%
order_up_to	raw	3	1.5	454.8281	76.1%	3.2%
order_up_to	raw	7	1.0	216.0284	91.6%	0.8%
order_up_to	raw	7	1.25	314.9469	84.5%	0.6%
order_up_to	raw	7	1.5	445.4677	76.3%	0.6%
order_up_to	selected_recovery	1	1.0	472.4809	80.0%	46.4%
order_up_to	selected_recovery	1	1.25	534.1117	75.1%	37.1%
order_up_to	selected_recovery	1	1.5	626.0210	69.5%	30.1%
order_up_to	selected_recovery	3	1.0	232.0698	92.1%	7.5%

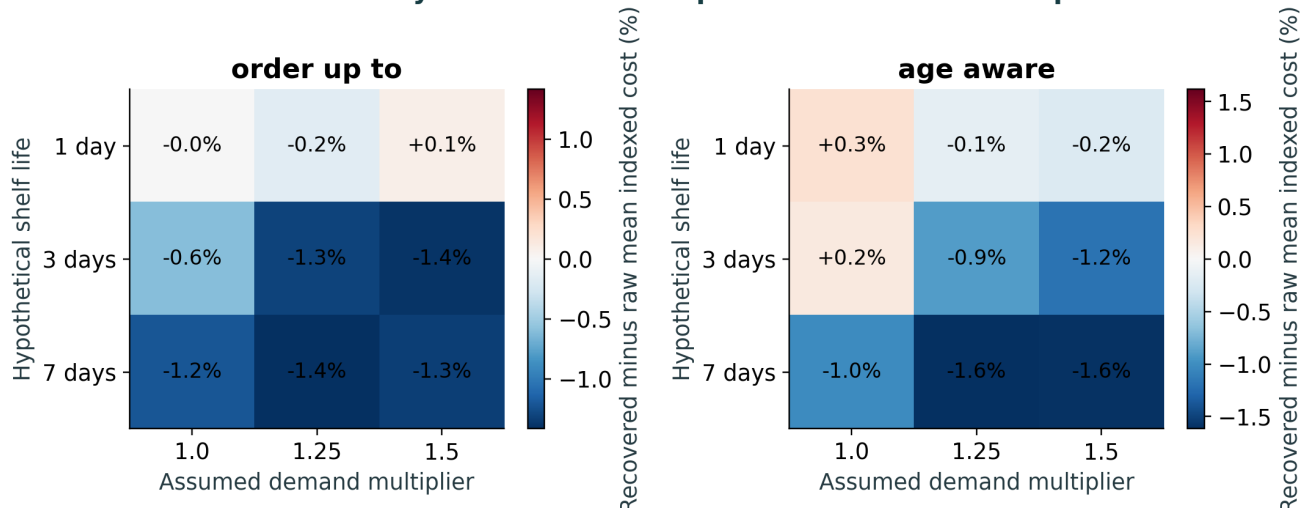
Calculator	History	Life	Demand×	Cost index	Fill	Expiry share
order_up_to	selected_recover	3	1.25	321.5021	85.0%	4.4%
order_up_to	selected_recover	3	1.5	448.5044	76.8%	3.3%
order_up_to	selected_recover	7	1.0	213.4033	92.1%	0.8%
order_up_to	selected_recover	7	1.25	310.4848	85.0%	0.6%
order_up_to	selected_recover	7	1.5	439.5487	76.9%	0.6%

Fresh retail exposes a service-waste trade-off



Independent calculator experiment, with 30 shared seeds per cell and demand multiplier 1. Shelf lives are hypothetical. Quantity and economic conventions differ from the core MILP experiment, so their absolute results are not pooled.

Demand recovery must earn its keep under different assumptions



Relative change from raw to recovered forecast history within each common-demand cell. Thirty seeds, shelf lives 1/3/7 days and demand multipliers 1/1.25/1.5. Negative values mean lower assumed indexed cost. They are not measured profit or purchasing savings.

8. M5 versus FreshRetailNet: a short descriptive comparison

Both panels contain 30 series, but their markets, products, dates, horizons, quantity units and economics differ. M5 supplies a long daily-unit sales hierarchy. FreshRetailNet supplies 90 training days of globally normalized sales with real hourly availability annotations. M5's matched agent runs last 14 days; the new runs last seven. Absolute cost levels cannot therefore form a meaningful commercial ranking.

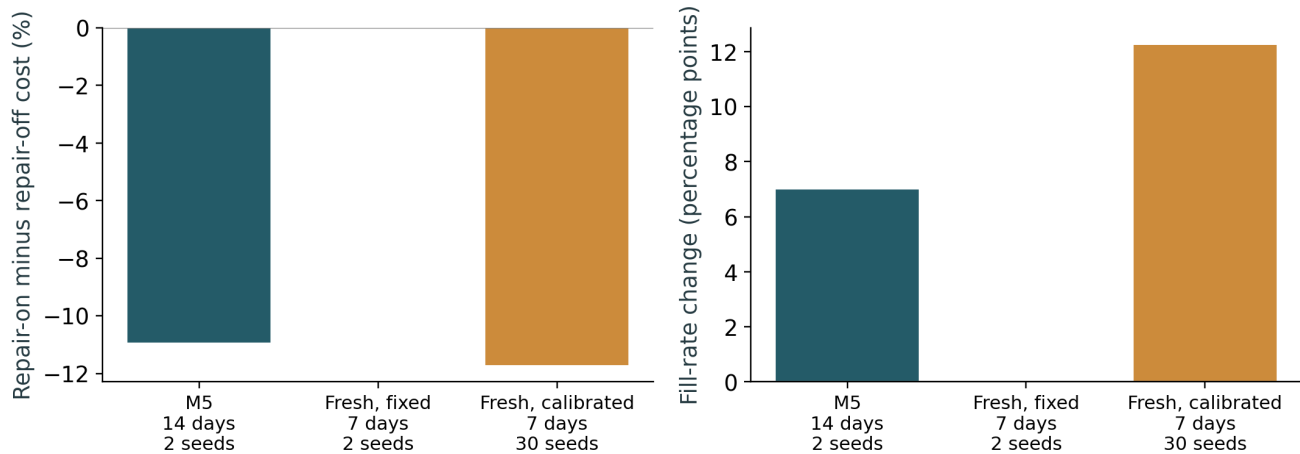
The compact comparison reports separately estimated within-dataset contrasts. In M5, parser and LLM cost and fill were identical in all 16 matched pairs. Record repair under the injected field fault reduced mean cost from 894.30 to 796.51, or 10.93%, and raised fill from 89% to 96%, or seven points. The corresponding FreshRetailNet contrasts appear below. These are descriptive engineering effects, not an estimate of what changing datasets does to a retailer's profit.

FreshRetailNet adds an information question that M5 cannot identify directly: a correct record can show constrained availability without showing how much demand was lost. That supports demand-visibility and perishability controls. It does not turn the language model into a better numerical forecaster. Additional semantic value would need fair, independently held-out tests of difficult supplier and exception language.

Within-dataset inventory-repair contrasts

Dataset	Scenario	Days	Cost change	Fill change(pp)	Identical parser/LLM pairs
M5	Normal	14	0.00%	0.00	4/4
M5	Inventory-field fault	14	-10.93%	7.00	4/4
M5	Stale feed	14	0.00%	0.00	4/4
M5	Capacity cut	14	0.00%	0.00	4/4
FreshRetailNet (fixed gate)	Normal	7	0.00%	0.00	4/4
FreshRetailNet (fixed gate)	Inventory-field fault	7	0.00%	0.00	4/4
FreshRetailNet (fixed gate)	Stale feed	7	0.00%	0.00	4/4
FreshRetailNet (fixed gate)	Capacity cut	7	0.00%	0.00	4/4
FreshRetailNet (calibrated gate)	Normal	7	0.00%	0.00	60/60
FreshRetailNet (calibrated gate)	Inventory-field fault	7	-11.71%	12.25	60/60
FreshRetailNet (calibrated gate)	Stale feed	7	0.00%	0.00	60/60
FreshRetailNet (calibrated gate)	Capacity cut	7	0.00%	0.00	60/60

Two information environments; two separately estimated effects



Separate within-dataset repair effects under an inventory-field fault. M5 uses 14 days and FreshRetailNet seven; their quantity units, expiry assumptions and economics differ. The calibrated result is an exploratory reused-holdout follow-up with 30 simulator seeds; the other bars have two. These bars do not estimate a commercial return or the causal effect of changing datasets.

9. What the results mean for a business

The business case supported by this evidence is for a controlled replenishment process. A retailer needs dependable stock evidence, a clearly labelled estimate when availability hides demand, typed supplier rules and checks before spending. An LLM can participate inside that process. The experiments have not established that it should replace the conventional numerical tools or a capable rules engine.

Useful language tasks remain plausible: translating supplier clauses into checkable fields, requesting a permitted exception tool, and producing a decision packet connected to its sources. The study demonstrates bounded feasibility for a controlled synthetic grammar. It does not measure planner time savings, human trust, unfamiliar natural-contract performance or production ROI. Those are distinct investment tests.

A sensible deployment starts in shadow mode with actual inventories and supplier documents, then moves to reviewed orders and spending caps. Compare exception-resolution quality and delay against the existing system. Expand autonomy only where the measured additional benefit covers error, oversight, hosting and integration costs. Where a capable parser gives the same business outcome more cheaply, it remains a reasonable default.

Answers to the retained research questions

Question	Completed answer	Scope limit
RQ1: performance	Conventional policies and matched parser/LLM outcomes are reported for each core scenario.	All single-agent and free-form architectures remain unexecuted.
RQ2: evidence gate	M5 detects known faults; FreshRetailNet exposes spending-cap portability and delay costs.	Effects depend on thresholds and simulator assumptions.
RQ3: censored demand	Recovery is measurable on masking; final forecast benefit is mixed; default service/expiry trade-off is small.	Natural unserved demand has no identified target.
RQ4: availability vs error	The explicit stockout-gate ablation shows a valid observation need not justify an integrity hold.	The naive comparator is a mechanism test, not a capable production parser.
RQ5: semantic value	Cost and fill match the capable parser on the disclosed synthetic grammar.	Unfamiliar, conflicting natural documents are not tested.
RQ6: agent organization	Typed outputs and independent checks are operationally audited.	No head-to-head test isolates additional agent roles.
RQ7: trace quality	Source-linked records, physical balances and attempts are machine-audited.	Human diagnosis and trust were not measured.
RQ8: generalizability	Within-dataset effects and scale-transfer failure are compared descriptively.	One small panel and origin cannot establish retailer-wide external validity.

Hypotheses: descriptive evidence status, without confirmatory tests

Hypothesis	Status	Reason
H1	Descriptively consistent	Matched grammar outcomes show no additional LLM cost/fill advantage; this is not a noninferiority test.

Hypothesis	Status	Reason
H2	Mechanism verified; incremental benefit bounded	Detected evidence faults trigger holds or verified repair. Conventional MILP already has zero actual rule violations; cap portability restricts useful execution.
H3	Mixed and conditional	Recovery selection, forecast accuracy and service-expiry trade-offs differ; natural demand recovery is not proved.
H4	Supported within the targeted ablation	Holding on valid stockout observations worsens simulated service against the censoring-aware rule.
H5	Supported on templates; remaining clause untested	The capable parser matches the LLM. Gains on unfamiliar documents were not evaluated.
H6	Untested as a comparative hypothesis	No free-form multi-agent baseline isolates typed-state or specialization effects.
H7	Untested	No participant study measures fault diagnosis, acceptance or trust.
H8	Conditional engineering evidence	Known-fault controls are useful, but the transferred cap fails until a separately calibrated exploratory follow-up.

10. Audit, limitations and reproducibility

The independent audit reads run folders, decision objects, grounding outputs and model request/response records. It verifies source hashes and event chains, matched comparisons, inventory-repair lineage and exclusion of future or evaluator-only information. Exact requests are written before HTTP calls. Development failures, setup receipts and any partial directories remain in the evidence archive.

The primary registered study audit captured 36 physical API attempts and 37506 tokens. There were 0 client or schema error events and 1269 committed true-constraint violations. Instruction-compliant selector responses and semantic selector errors are reported separately below; successful final safety is not a substitute for reliable tool choice. The exploratory follow-up has its own audit and operating counts in its preceding section.

Thirty seeds describe simulator variability conditional on one panel, one origin and assumed economics. The live factorial has two seeds and reused cached evidence. The short forecasting holdout and its conditionally stocked subset restrict generalization. No confirmatory p-value, noninferiority claim, participant result or retailer return is asserted.

The wider dissertation proposes additional B1-B11 architectures, a semantic difficulty ladder, free-form multi-agent alternatives, prompt-injection testing and a human study. Those remain unevaluated unless their own saved executions exist. This bounded study completes the registered controls and specified F1-F6 subtests; it does not silently promote the whole proposed research program into a completed result.

Data attribution: Dingdong-Inc, FreshRetailNet-50K, CC BY 4.0; Wang et al. (2025), pinned dataset revision 08c1fab7f9257bc73679d415d65d644165d351d4. M5 retains its competition provenance. The supplied literature review and references are preserved in the updated Word dissertation.

The overall audit's 1,269 requested-plan violations are concentrated in naive numerical B1/B2. The live LLM arms have 0; the conventional MILP B3 also has zero. An operational architecture should report feasibility and service together rather than claim a language-specific safety benefit from this difference.

Primary registered study audit: distinct operational counts

Measure	Recorded count
Completed runs	392
Decision days	2744
Conventional core runs	360
Matched controller runs	32
Fresh model attempts	36
Fresh model responses	36
Recorded model tokens	37506
Client or schema errors	0
Fresh numerical-extraction responses	24
Fresh recovery selections	12

Measure	Recorded count
Violated active-rule checks in requested plans	1269
Held decision days	814
Instruction-compliant recovery selections	6
Incorrect recovery selections	6
Matched parser-LLM pairs	16
Pairs with identical primary outcomes	16
Pairs with identical daily actions	16
Verified physical stock balances	2744
Forbidden model-context findings	0

Source grounding coverage; full columns in the audit CSV

Configuration	Grounding attempted	Exact full source sets	State-gated days	False clauses	Omitted clauses
numerical	2340	2340	180	0	0
Parser	44	44	12	0	0
Parser + repair	50	50	6	0	0
LLM	44	44	12	0	0
LLM + repair	50	50	6	0	0

Appendix A. Artifact guide and exact run-level results

The accompanying files preserve every completed individual run. Numerical, live-agent and independent-policy CSVs have 360, 32 and 1,140 rows respectively. Forecast files contain series/day predictions, observable targets and flags used only for scoring. Artificial-mask files retain each hidden target. Audit files contain model calls, grounding, selector and decision records.

The report directory contains FreshRetailNet_numerical_all_runs.csv, FreshRetailNet_agent_all_runs.csv and FreshRetailNet_policy_sensitivity_all_runs.csv, as well as their descriptive means and matched contrasts. M5_FreshRetailNet_descriptive_comparison.csv records the compatible relative comparisons. The full evidence archive adds raw objects, saved models, frozen code, source receipts and failed attempts.

Selected computational panel

Series	Stratum	Train mean sales	Train stockout share
FRN_0247@S0020	2	0.38677	22.28%
FRN_0363@S0021	2	0.39774	26.31%
FRN_0210@S0054	2	0.33565	23.69%
FRN_0218@S0137	1	0.90758	17.34%
FRN_0635@S0156	1	0.81226	18.25%
FRN_0422@S0173	1	0.93710	20.26%
FRN_0738@S0181	2	0.28823	23.69%
FRN_0190@S0191	0	0.83548	8.87%
FRN_0295@S0202	1	0.37645	18.95%
FRN_0653@S0202	0	0.35194	14.82%
FRN_0635@S0219	1	0.42403	17.84%
FRN_0473@S0231	0	0.51613	15.02%
FRN_0802@S0259	0	0.84839	15.93%
FRN_0769@S0298	0	0.70000	12.80%
FRN_0843@S0300	2	0.54355	32.66%
FRN_0682@S0301	0	0.96290	9.78%
FRN_0834@S0334	2	0.51290	60.38%
FRN_0802@S0355	1	0.72097	16.43%
FRN_0810@S0410	0	0.48903	9.98%
FRN_0715@S0413	2	0.35452	23.69%
FRN_0650@S0423	0	0.45597	14.72%
FRN_0729@S0474	0	0.49355	10.99%
FRN_0218@S0482	0	0.66452	11.39%
FRN_0620@S0484	1	0.60774	19.35%

Series	Stratum	Train mean sales	Train stockout share
FRN_0577@S0521	1	0.42242	17.34%
FRN_0834@S0649	2	0.92419	45.77%
FRN_0560@S0805	1	0.57726	17.74%
FRN_0074@S0851	2	0.55387	22.18%
FRN_0402@S0861	1	0.56629	19.56%
FRN_0379@S0877	2	0.54516	55.44%

Appendix B. Core individual runs — Normal

All 30 seeds per policy

Policy	Seed	Cost	Fill	Stockout	Held	Violations
B1	0	17,735.329	78.80%	27.62%	0	7
B1	1	16,579.451	81.03%	27.62%	0	6
B1	2	17,420.022	79.04%	28.57%	0	7
B1	3	18,125.015	77.93%	30.00%	0	6
B1	4	19,701.915	74.99%	30.95%	0	7
B1	5	17,780.512	77.92%	29.05%	0	7
B1	6	19,138.125	75.20%	31.90%	0	7
B1	7	18,583.141	77.28%	29.05%	0	7
B1	8	17,731.543	78.18%	29.05%	0	7
B1	9	20,937.615	70.63%	35.71%	0	7
B1	10	16,176.249	81.99%	23.33%	0	7
B1	11	19,119.515	75.78%	31.43%	0	7
B1	12	19,894.785	73.82%	30.00%	0	7
B1	13	18,242.236	77.23%	29.52%	0	7
B1	14	19,224.607	76.77%	30.00%	0	7
B1	15	18,838.073	76.21%	26.67%	0	8
B1	16	18,331.092	76.75%	26.67%	0	7
B1	17	17,990.827	78.62%	31.43%	0	7
B1	18	18,288.650	77.44%	30.95%	0	7
B1	19	16,198.008	82.75%	24.76%	0	7
B1	20	18,173.944	78.53%	26.19%	0	7
B1	21	18,464.016	76.82%	27.14%	0	7
B1	22	17,714.833	79.14%	25.24%	0	7
B1	23	18,081.225	77.84%	29.52%	0	7
B1	24	17,709.263	78.39%	29.52%	0	7
B1	25	18,315.159	76.78%	26.67%	0	8
B1	26	17,002.835	81.65%	23.81%	0	7
B1	27	18,872.682	77.91%	29.52%	0	7
B1	28	19,199.945	75.61%	33.81%	0	7
B1	29	19,139.064	75.72%	28.10%	0	7
B2	0	18,714.549	73.04%	32.86%	0	6

Policy	Seed	Cost	Fill	Stockout	Held	Violations
B2	1	19,450.479	70.16%	36.67%	0	4
B2	2	18,221.380	71.89%	33.33%	0	7
B2	3	20,319.804	68.74%	35.71%	0	6
B2	4	20,524.152	68.68%	35.24%	0	7
B2	5	20,375.280	67.35%	37.14%	0	5
B2	6	20,247.988	69.68%	36.19%	0	7
B2	7	20,150.511	68.26%	35.71%	0	7
B2	8	20,921.230	65.55%	38.10%	0	7
B2	9	22,216.301	64.51%	38.10%	0	7
B2	10	18,691.531	72.37%	33.81%	0	6
B2	11	19,969.856	68.91%	35.71%	0	4
B2	12	20,875.138	67.17%	38.57%	0	7
B2	13	19,058.313	71.17%	37.14%	0	5
B2	14	21,143.615	67.12%	36.67%	0	6
B2	15	20,538.898	66.91%	37.62%	0	7
B2	16	19,760.069	69.79%	32.38%	0	6
B2	17	17,376.287	75.41%	33.81%	0	7
B2	18	19,133.461	69.75%	35.24%	0	6
B2	19	17,718.816	74.84%	30.95%	0	6
B2	20	20,228.331	69.08%	38.57%	0	7
B2	21	20,047.329	68.73%	37.14%	0	6
B2	22	19,655.185	70.05%	34.76%	0	6
B2	23	18,658.486	71.88%	33.81%	0	7
B2	24	20,149.638	68.19%	39.05%	0	7
B2	25	18,722.456	71.34%	32.86%	0	6
B2	26	19,227.461	71.22%	34.76%	0	7
B2	27	18,135.282	73.70%	34.29%	0	7
B2	28	20,076.943	68.79%	37.62%	0	7
B2	29	20,254.479	67.94%	35.24%	0	7
B3	0	18,834.188	85.17%	19.52%	0	0
B3	1	18,489.906	86.17%	17.14%	0	0
B3	2	18,667.147	84.51%	17.62%	0	0
B3	3	19,054.028	83.53%	24.29%	0	0
B3	4	20,285.256	81.83%	26.19%	0	0

Policy	Seed	Cost	Fill	Stockout	Held	Violations
B3	5	19,219.266	84.43%	20.95%	0	0
B3	6	17,918.294	88.27%	16.19%	0	0
B3	7	18,421.132	85.98%	18.57%	0	0
B3	8	18,481.642	85.25%	20.95%	0	0
B3	9	20,756.792	80.59%	27.62%	0	0
B3	10	18,482.335	86.34%	18.10%	0	0
B3	11	20,111.100	81.98%	21.43%	0	0
B3	12	20,453.425	82.34%	21.90%	0	0
B3	13	17,996.392	85.59%	19.52%	0	0
B3	14	19,358.248	83.05%	22.86%	0	0
B3	15	17,988.876	86.53%	15.71%	0	0
B3	16	18,656.536	84.80%	19.05%	0	0
B3	17	17,781.322	87.27%	20.00%	0	0
B3	18	20,510.611	80.13%	24.76%	0	0
B3	19	16,054.654	91.33%	16.67%	0	0
B3	20	18,379.175	85.74%	20.48%	0	0
B3	21	19,573.899	83.19%	23.81%	0	0
B3	22	18,493.814	85.83%	17.62%	0	0
B3	23	17,458.419	88.26%	17.14%	0	0
B3	24	18,711.980	85.16%	20.48%	0	0
B3	25	18,223.855	86.36%	15.71%	0	0
B3	26	18,638.899	85.59%	20.00%	0	0
B3	27	18,396.860	85.75%	18.57%	0	0
B3	28	18,730.647	85.91%	20.48%	0	0
B3	29	19,692.291	83.95%	19.05%	0	0
B4	0	24,488.115	42.27%	60.95%	7	0
B4	1	25,521.012	39.25%	60.95%	7	0
B4	2	23,654.760	43.89%	56.67%	7	0
B4	3	23,251.410	45.19%	59.05%	7	0
B4	4	24,019.237	44.22%	55.24%	7	0
B4	5	24,508.003	41.92%	60.95%	7	0
B4	6	22,696.454	47.58%	52.86%	7	0
B4	7	23,110.907	45.83%	54.76%	7	0
B4	8	25,396.775	39.78%	60.95%	7	0

Policy	Seed	Cost	Fill	Stockout	Held	Violations
B4	9	26,169.138	38.52%	61.90%	7	0
B4	10	23,095.977	45.08%	56.19%	7	0
B4	11	23,592.960	45.42%	56.19%	7	0
B4	12	25,421.616	40.38%	60.95%	7	0
B4	13	22,258.569	47.43%	53.33%	7	0
B4	14	25,075.476	40.86%	60.95%	7	0
B4	15	23,274.089	45.64%	55.24%	7	0
B4	16	24,461.748	42.52%	57.62%	7	0
B4	17	24,095.772	43.55%	56.67%	7	0
B4	18	23,666.492	44.01%	56.67%	7	0
B4	19	22,742.703	45.99%	55.24%	7	0
B4	20	24,289.215	42.78%	61.43%	7	0
B4	21	24,624.152	42.41%	59.05%	7	0
B4	22	22,626.153	46.72%	55.24%	7	0
B4	23	23,949.690	43.01%	59.52%	7	0
B4	24	22,468.061	47.18%	57.62%	7	0
B4	25	24,722.401	41.14%	55.71%	7	0
B4	26	23,381.589	45.43%	56.67%	7	0
B4	27	22,401.393	47.23%	53.81%	7	0
B4	28	27,026.395	36.69%	61.90%	7	0
B4	29	24,404.020	42.93%	57.62%	7	0

Appendix B. Core individual runs — Inventory-field fault

All 30 seeds per policy

Policy	Seed	Cost	Fill	Stockout	Held	Violations
B1	0	17,611.796	80.89%	23.81%	0	9
B1	1	17,589.209	80.09%	26.67%	0	7
B1	2	18,685.214	78.41%	25.71%	0	10
B1	3	17,961.443	80.07%	29.05%	0	9
B1	4	19,659.989	76.54%	28.10%	0	10
B1	5	17,568.577	80.82%	29.05%	0	8
B1	6	18,918.969	76.56%	29.52%	0	10
B1	7	18,806.993	78.62%	26.67%	0	10
B1	8	19,059.460	75.85%	31.90%	0	9
B1	9	21,289.694	71.65%	33.33%	0	8
B1	10	17,013.616	81.32%	23.33%	0	8
B1	11	18,275.851	78.94%	29.52%	0	9
B1	12	19,382.016	77.21%	28.10%	0	9
B1	13	18,608.672	77.57%	28.57%	0	9
B1	14	19,359.835	79.00%	26.19%	0	8
B1	15	18,303.688	79.36%	24.76%	0	12
B1	16	19,139.488	75.91%	27.62%	0	9
B1	17	19,185.711	77.25%	30.95%	0	8
B1	18	18,520.027	78.35%	28.10%	0	7
B1	19	17,086.143	82.92%	20.48%	0	9
B1	20	19,148.857	79.62%	24.76%	0	8
B1	21	17,851.136	79.99%	25.24%	0	8
B1	22	17,942.610	79.55%	24.29%	0	10
B1	23	19,336.243	76.16%	28.57%	0	8
B1	24	17,585.124	80.62%	25.71%	0	9
B1	25	19,130.153	77.19%	25.71%	0	10
B1	26	16,293.325	85.01%	20.00%	0	8
B1	27	18,338.343	80.33%	25.24%	0	9
B1	28	20,166.117	75.06%	30.95%	0	8
B1	29	20,488.995	73.67%	30.95%	0	10
B2	0	19,224.390	73.48%	31.43%	0	6

Policy	Seed	Cost	Fill	Stockout	Held	Violations
B2	1	18,885.377	73.69%	34.29%	0	7
B2	2	18,694.716	73.38%	30.95%	0	7
B2	3	19,100.233	73.24%	32.86%	0	8
B2	4	19,636.017	72.01%	30.95%	0	8
B2	5	19,643.144	71.80%	33.81%	0	6
B2	6	19,168.154	72.87%	32.86%	0	7
B2	7	19,560.190	72.15%	33.33%	0	8
B2	8	20,989.762	66.70%	36.67%	0	8
B2	9	21,858.340	66.05%	35.71%	0	8
B2	10	18,715.354	72.90%	33.81%	0	7
B2	11	18,352.859	73.56%	30.00%	0	5
B2	12	19,607.219	72.28%	33.33%	0	6
B2	13	18,729.918	73.90%	31.90%	0	7
B2	14	20,543.536	70.27%	34.29%	0	6
B2	15	18,394.891	74.70%	30.48%	0	7
B2	16	19,953.326	71.05%	31.43%	0	8
B2	17	16,720.617	79.08%	29.52%	0	5
B2	18	19,900.279	69.34%	32.86%	0	6
B2	19	16,832.209	79.17%	25.71%	0	7
B2	20	19,189.160	74.64%	32.38%	0	7
B2	21	19,267.769	73.14%	33.81%	0	6
B2	22	18,473.638	74.54%	30.48%	0	7
B2	23	19,915.621	70.34%	33.81%	0	8
B2	24	19,043.478	72.56%	35.71%	0	7
B2	25	18,339.779	75.53%	26.19%	0	7
B2	26	18,807.756	74.88%	30.48%	0	6
B2	27	17,459.968	77.19%	29.05%	0	5
B2	28	19,465.057	72.23%	32.86%	0	5
B2	29	20,701.076	69.50%	32.86%	0	9
B3	0	19,159.301	84.76%	20.48%	0	0
B3	1	18,207.028	86.83%	17.62%	0	0
B3	2	18,910.859	84.39%	17.14%	0	0
B3	3	18,824.738	84.91%	23.81%	0	0
B3	4	20,468.676	82.03%	24.76%	0	0

Policy	Seed	Cost	Fill	Stockout	Held	Violations
B3	5	19,807.642	83.44%	21.43%	0	0
B3	6	18,098.312	88.55%	14.76%	0	0
B3	7	19,213.589	85.01%	18.57%	0	0
B3	8	19,358.964	83.92%	21.90%	0	0
B3	9	20,934.487	80.31%	26.19%	0	0
B3	10	18,774.688	86.04%	17.62%	0	0
B3	11	20,378.474	82.13%	21.43%	0	0
B3	12	19,802.567	84.18%	21.43%	0	0
B3	13	18,029.206	85.94%	20.00%	0	0
B3	14	20,067.941	82.05%	24.29%	0	0
B3	15	18,315.320	86.83%	15.24%	0	0
B3	16	18,836.141	84.45%	19.52%	0	0
B3	17	17,830.934	87.53%	20.48%	0	0
B3	18	19,853.349	81.93%	22.86%	0	0
B3	19	16,183.520	91.47%	15.71%	0	0
B3	20	18,550.147	85.44%	20.00%	0	0
B3	21	19,678.908	83.43%	23.33%	0	0
B3	22	18,593.378	86.19%	18.10%	0	0
B3	23	18,055.491	87.26%	18.10%	0	0
B3	24	18,953.864	85.31%	20.48%	0	0
B3	25	18,434.459	86.27%	15.71%	0	0
B3	26	18,888.526	86.04%	19.52%	0	0
B3	27	18,410.566	85.89%	18.10%	0	0
B3	28	19,522.936	84.32%	22.38%	0	0
B3	29	20,659.723	82.44%	20.00%	0	0
B4	0	24,488.115	42.27%	60.95%	7	0
B4	1	25,521.012	39.25%	60.95%	7	0
B4	2	23,654.760	43.89%	56.67%	7	0
B4	3	23,251.410	45.19%	59.05%	7	0
B4	4	24,019.237	44.22%	55.24%	7	0
B4	5	24,508.003	41.92%	60.95%	7	0
B4	6	22,696.454	47.58%	52.86%	7	0
B4	7	23,110.907	45.83%	54.76%	7	0
B4	8	25,396.775	39.78%	60.95%	7	0

Policy	Seed	Cost	Fill	Stockout	Held	Violations
B4	9	26,169.138	38.52%	61.90%	7	0
B4	10	23,095.977	45.08%	56.19%	7	0
B4	11	23,592.960	45.42%	56.19%	7	0
B4	12	25,421.616	40.38%	60.95%	7	0
B4	13	22,258.569	47.43%	53.33%	7	0
B4	14	25,075.476	40.86%	60.95%	7	0
B4	15	23,274.089	45.64%	55.24%	7	0
B4	16	24,461.748	42.52%	57.62%	7	0
B4	17	24,095.772	43.55%	56.67%	7	0
B4	18	23,666.492	44.01%	56.67%	7	0
B4	19	22,742.703	45.99%	55.24%	7	0
B4	20	24,289.215	42.78%	61.43%	7	0
B4	21	24,624.152	42.41%	59.05%	7	0
B4	22	22,626.153	46.72%	55.24%	7	0
B4	23	23,949.690	43.01%	59.52%	7	0
B4	24	22,468.061	47.18%	57.62%	7	0
B4	25	24,722.401	41.14%	55.71%	7	0
B4	26	23,381.589	45.43%	56.67%	7	0
B4	27	22,401.393	47.23%	53.81%	7	0
B4	28	27,026.395	36.69%	61.90%	7	0
B4	29	24,404.020	42.93%	57.62%	7	0

Appendix B. Core individual runs — Stale feed

All 30 seeds per policy

Policy	Seed	Cost	Fill	Stockout	Held	Violations
B1	0	19,512.420	76.23%	29.52%	0	7
B1	1	17,789.008	78.99%	29.52%	0	6
B1	2	18,464.022	76.88%	27.62%	0	7
B1	3	18,885.500	77.44%	29.52%	0	7
B1	4	21,575.320	71.89%	34.29%	0	7
B1	5	18,314.971	77.65%	30.95%	0	7
B1	6	20,251.288	72.72%	31.43%	0	7
B1	7	19,261.653	76.31%	29.52%	0	7
B1	8	20,248.948	72.98%	33.33%	0	7
B1	9	21,721.574	70.06%	36.67%	0	7
B1	10	17,104.856	80.92%	24.76%	0	7
B1	11	19,604.691	74.55%	33.33%	0	7
B1	12	22,158.547	69.08%	32.38%	0	7
B1	13	19,903.064	73.67%	31.90%	0	7
B1	14	20,886.125	74.43%	29.52%	0	7
B1	15	19,400.510	75.49%	30.00%	0	8
B1	16	20,417.276	71.54%	30.00%	0	5
B1	17	19,236.356	77.49%	30.48%	0	7
B1	18	19,505.589	75.26%	32.38%	0	7
B1	19	17,243.807	81.15%	26.19%	0	7
B1	20	18,857.179	77.74%	26.19%	0	7
B1	21	19,556.668	75.18%	29.52%	0	7
B1	22	18,601.854	78.16%	26.67%	0	7
B1	23	19,337.286	74.87%	31.90%	0	7
B1	24	19,385.310	74.23%	32.38%	0	7
B1	25	18,607.195	77.44%	26.67%	0	8
B1	26	18,492.408	78.89%	24.29%	0	7
B1	27	19,230.189	77.53%	28.10%	0	7
B1	28	19,309.966	76.31%	30.95%	0	7
B1	29	20,568.612	74.55%	30.48%	0	7
B2	0	23,223.296	65.09%	37.14%	0	7

Policy	Seed	Cost	Fill	Stockout	Held	Violations
B2	1	21,682.569	66.33%	39.52%	0	6
B2	2	21,849.802	64.26%	40.00%	0	7
B2	3	22,811.511	64.25%	40.48%	0	8
B2	4	22,376.619	65.28%	39.52%	0	7
B2	5	21,317.351	64.60%	40.48%	0	3
B2	6	21,716.401	66.17%	39.52%	0	8
B2	7	23,160.512	62.08%	40.48%	0	7
B2	8	25,032.830	59.20%	43.33%	0	6
B2	9	24,327.803	63.19%	40.95%	0	7
B2	10	21,270.958	67.97%	38.57%	0	6
B2	11	22,663.677	62.45%	40.48%	0	3
B2	12	24,527.174	60.83%	43.33%	0	7
B2	13	21,185.771	67.82%	36.67%	0	6
B2	14	24,928.814	62.76%	41.43%	0	7
B2	15	22,112.482	65.63%	40.48%	0	7
B2	16	22,560.230	62.85%	38.57%	0	6
B2	17	20,873.872	71.40%	38.10%	0	7
B2	18	22,333.899	63.95%	37.62%	0	4
B2	19	20,311.691	69.85%	34.76%	0	6
B2	20	21,149.794	69.26%	38.10%	0	7
B2	21	24,461.539	60.01%	45.24%	0	6
B2	22	21,273.712	67.88%	37.14%	0	5
B2	23	21,481.098	65.38%	42.38%	0	7
B2	24	22,977.143	62.41%	42.86%	0	6
B2	25	20,330.263	71.55%	33.33%	0	7
B2	26	22,702.083	65.02%	38.57%	0	6
B2	27	20,486.263	71.82%	34.29%	0	7
B2	28	23,435.644	64.00%	40.48%	0	7
B2	29	22,755.512	66.08%	36.67%	0	7
B3	0	20,299.390	81.93%	23.33%	0	0
B3	1	19,181.198	84.36%	19.52%	0	0
B3	2	19,816.853	81.88%	20.48%	0	0
B3	3	20,192.821	81.83%	25.71%	0	0
B3	4	20,119.272	81.46%	27.62%	0	0

Policy	Seed	Cost	Fill	Stockout	Held	Violations
B3	5	18,839.755	84.72%	21.43%	0	0
B3	6	19,174.011	85.43%	18.57%	0	0
B3	7	18,986.265	85.21%	19.05%	0	0
B3	8	20,914.570	80.35%	24.29%	0	0
B3	9	20,949.736	80.29%	28.10%	0	0
B3	10	18,760.641	84.71%	20.00%	0	0
B3	11	20,951.077	81.03%	21.90%	0	0
B3	12	21,062.438	80.61%	24.29%	0	0
B3	13	18,908.825	83.84%	21.43%	0	0
B3	14	20,392.447	81.27%	25.71%	0	0
B3	15	18,900.140	85.14%	16.19%	0	0
B3	16	19,650.705	82.26%	19.52%	0	0
B3	17	18,012.580	86.77%	21.43%	0	0
B3	18	21,277.099	78.80%	25.24%	0	0
B3	19	17,616.039	88.01%	18.10%	0	0
B3	20	18,487.261	85.48%	20.48%	0	0
B3	21	20,091.646	82.37%	24.76%	0	0
B3	22	19,692.357	83.04%	19.52%	0	0
B3	23	18,598.392	85.74%	19.05%	0	0
B3	24	19,229.893	83.89%	21.43%	0	0
B3	25	19,035.589	84.51%	17.14%	0	0
B3	26	20,289.185	81.74%	23.81%	0	0
B3	27	18,815.932	85.06%	18.57%	0	0
B3	28	19,332.990	84.30%	23.33%	0	0
B3	29	20,631.400	82.26%	20.95%	0	0
B4	0	24,488.115	42.27%	60.95%	7	0
B4	1	25,521.012	39.25%	60.95%	7	0
B4	2	23,654.760	43.89%	56.67%	7	0
B4	3	23,251.410	45.19%	59.05%	7	0
B4	4	24,019.237	44.22%	55.24%	7	0
B4	5	24,508.003	41.92%	60.95%	7	0
B4	6	22,696.454	47.58%	52.86%	7	0
B4	7	23,110.907	45.83%	54.76%	7	0
B4	8	25,396.775	39.78%	60.95%	7	0

Policy	Seed	Cost	Fill	Stockout	Held	Violations
B4	9	26,169.138	38.52%	61.90%	7	0
B4	10	23,095.977	45.08%	56.19%	7	0
B4	11	23,592.960	45.42%	56.19%	7	0
B4	12	25,421.616	40.38%	60.95%	7	0
B4	13	22,258.569	47.43%	53.33%	7	0
B4	14	25,075.476	40.86%	60.95%	7	0
B4	15	23,274.089	45.64%	55.24%	7	0
B4	16	24,461.748	42.52%	57.62%	7	0
B4	17	24,095.772	43.55%	56.67%	7	0
B4	18	23,666.492	44.01%	56.67%	7	0
B4	19	22,742.703	45.99%	55.24%	7	0
B4	20	24,289.215	42.78%	61.43%	7	0
B4	21	24,624.152	42.41%	59.05%	7	0
B4	22	22,626.153	46.72%	55.24%	7	0
B4	23	23,949.690	43.01%	59.52%	7	0
B4	24	22,468.061	47.18%	57.62%	7	0
B4	25	24,722.401	41.14%	55.71%	7	0
B4	26	23,381.589	45.43%	56.67%	7	0
B4	27	22,401.393	47.23%	53.81%	7	0
B4	28	27,026.395	36.69%	61.90%	7	0
B4	29	24,404.020	42.93%	57.62%	7	0

Appendix C. Live-agent individual runs

All 32 live factorial runs

Arm	Scenario	Seed	Cost	Fill	Held	Violations
LLM	Capacity cut	13	20,589.055	59.41%	2	0
LLM	Capacity cut	29	22,556.878	55.46%	2	0
LLM	Inventory-field fault	13	22,258.569	47.43%	7	0
LLM	Inventory-field fault	29	24,404.020	42.93%	7	0
LLM	Stale feed	13	22,258.569	47.43%	7	0
LLM	Stale feed	29	24,404.020	42.93%	7	0
LLM	Normal	13	22,258.569	47.43%	7	0
LLM	Normal	29	24,404.020	42.93%	7	0
LLM + repair	Capacity cut	13	20,589.055	59.41%	2	0
LLM + repair	Capacity cut	29	22,556.878	55.46%	2	0
LLM + repair	Inventory-field fault	13	22,258.569	47.43%	7	0
LLM + repair	Inventory-field fault	29	24,404.020	42.93%	7	0
LLM + repair	Stale feed	13	22,258.569	47.43%	7	0
LLM + repair	Stale feed	29	24,404.020	42.93%	7	0
LLM + repair	Normal	13	22,258.569	47.43%	7	0
LLM + repair	Normal	29	24,404.020	42.93%	7	0
Parser	Capacity cut	13	20,589.055	59.41%	2	0
Parser	Capacity cut	29	22,556.878	55.46%	2	0
Parser	Inventory-field fault	13	22,258.569	47.43%	7	0
Parser	Inventory-field fault	29	24,404.020	42.93%	7	0
Parser	Stale feed	13	22,258.569	47.43%	7	0
Parser	Stale feed	29	24,404.020	42.93%	7	0
Parser	Normal	13	22,258.569	47.43%	7	0
Parser	Normal	29	24,404.020	42.93%	7	0
Parser + repair	Capacity cut	13	20,589.055	59.41%	2	0
Parser + repair	Capacity cut	29	22,556.878	55.46%	2	0
Parser + repair	Inventory-field fault	13	22,258.569	47.43%	7	0

Arm	Scenario	Seed	Cost	Fill	Held	Violations
Parser + repair	Inventory-field fault	29	24,404.020	42.93%	7	0
Parser + repair	Stale feed	13	22,258.569	47.43%	7	0
Parser + repair	Stale feed	29	24,404.020	42.93%	7	0
Parser + repair	Normal	13	22,258.569	47.43%	7	0
Parser + repair	Normal	29	24,404.020	42.93%	7	0